

ShORRT
(Short, all-Oral Regimens for Rifampicin-resistant Tuberculosis)
Research Package

TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom.

Study Protocol

Version 1.2 – 23 July 2024

Sponsor: University College London NHS Foundation Trust (UCLH)

Sponsor reference number: 154296

Funder: To be confirmed

IRAS Number: 317172

ISRCTN: 14005421

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Version	Date	Reason of change
0.0	15 June 2022	Adaptation to UK context. Initial draft.
0.1	9 August 2022	Updates following meeting of potential site contributors on 22 nd July. Expanded to include isoniazid-resistant TB and patients intolerant to first-line medications.
0.2	6 September 2022	Review with DM.
0.3	11 October 2022	Updates following study management group meeting
0.4	2 January 2023	Further updates based on RDS and PPI consultations; and for compatibility with UCLH JRO requirements
0.5	10 April 2023	Revised structure into work packages
0.6	12 July 2023	Updated Part B with input from work package leads
1.1	17 July 2024	Updated timelines
1.2	23 July 2024	Updated Sponsor details and format

DECLARATIONS

The undersigned confirm that the following protocol has been agreed and accepted and that the investigator agrees to conduct the study in compliance with the approved protocol and will adhere to the U.K. Policy Framework for Health and Social Care Research 2017 (3rd edition) (as amended thereafter), the EU General Data Protection Regulation (2016/679) and the UK Data Protection Act (2018), Sponsor SOPs and applicable Trust policies and legal frameworks.

I (investigator) agree to ensure that the confidential information contained in this document will not be used for any other purposes other than the evaluation or conduct of the research investigation without the prior written consent of the Sponsor.

I (investigator) agree to ensure that no research activity or recruitment will commence at participating research sites until the appropriate regulatory approvals and NHS confirmations of Capacity and Capability have been issued, and Sponsor green light confirmed.

I (investigator) also confirm that I will make the findings of the study publicly available through publication or other dissemination tools without any unnecessary delay and that an honest, accurate and transparent account of the study will be given. Any deviations from the study as planned in this protocol will be explained and reported accordingly.

Chief Investigator:

Signature:



Date 17 / 07 / 2024

Print Name (in full):David Moore.....

Position:Consultant in Infectious Diseases and Tropical Medicine

On behalf of the Study Sponsor:

Signature:



Date 23 / 07 / 2024

Print Name (in full): Pushpsen Joshi

Position: Research Governance Manager

STUDY SUMMARY

IDENTIFIERS	
IRAS Number	317172
REC Reference No.	TBC
Sponsor Reference No.	154296
Other research reference number(s) (if applicable)	N/A
Full (Scientific) title	TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom.
Health condition(s) or problem(s) studied	Tuberculosis
Study Type i.e. Cohort etc.	Observational cohort
Target sample size	460
STUDY TIMELINES	
Study Duration/length	3 years
Expected Start Date	02/09/2024
End of Study definition and anticipated date	29/12/2028
Key Study milestones	- HRA / IRAS Submission July 2024 - First patient recruited September 2024 - Last patient recruited September 2026 - Study end December 2028
FUNDING & OTHER	
Funding	TBC
Other support	
STORAGE of SAMPLES / DATA (if applicable)	
Human tissue samples	N/A
Data collected / Storage	Patient data for Work Packages 1, 2 and 4 will be collected onto a REDCap database housed on servers of the World Health Organisation (WHO). For Work Package 3, qualitative data and transcripts will be stored securely on servers of University College London (UCL Data Safe Haven; ISO27001 certified).
KEY STUDY CONTACTS	
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Funder(s)	TBC
Committees	Study Management Group: David Moore, Peter Crook, Tom Harrison, Gerry Davies, Onn Min Kon, Marc Lipman, Martin Dedicoat, Patrick Lillie, Padmasayee Papineni, Amber Arnold, Heinke Kunst, Effrossyni Gkrania-Klotsas, Emily Wise, Annabelle South, Sedona Sweeney, Duncan Barron

	Public/Patient Involvement (PPI) Advisory Group: Led by Duncan Barron; members TBC
Sub-contractors	St George's, University of London (Dr Peter Crook) University College London (Dr Annabelle South, Dr Conor Tweed) London School of Hygiene and Tropical Medicine (Dr Sedona Sweeney, Prof David Moore)

KEY ROLES AND RESPONSIBILITIES

SPONSOR: The sponsor is responsible for ensuring before a study begins that arrangements are in place for the research team to access resources and support to deliver the research as proposed and allocate responsibilities for the management, monitoring and reporting of the research. The Sponsor also must be satisfied there is agreement on appropriate arrangements to record, report and review significant developments as the research proceeds, and approve any modifications to the design.

FUNDER: The funder is the entity that will provide the funds (financial support) for the conduction of the study. Funders are expected to provide assistance to any enquiry, audit or investigation related to the funded work.

CHIEF INVESTIGATOR (CI): The person who takes overall responsibility for the design, conduct and reporting of a study. If the study involves researchers at more than one site, the CI takes on the primary responsibility irrespective of whether he/she is an investigator at any particular site.

The CI role is to complete and to ensure that all relevant regulatory approvals and confirmations of NHS Capacity and Capability are in place before the study begins. Ensure arrangements are in place for good study conduct, robust monitoring and reporting, including prompt reporting of incidents, this includes putting in place adequate training for study staff to conduct the study as per the protocol and relevant standards.

The Chief Investigator is responsible for submission of annual reports as required. The Chief Investigator will notify the REC and JRO of the end of the study (including the reasons for premature termination, where applicable). Within one year after the end of study, the Chief Investigator will submit a final report with the results, including any publications/abstracts to the REC and JRO.

PRINCIPAL INVESTIGATOR (PI): Individually or as leader of the researchers at a site; ensuring that the study is conducted as per the approved study protocol, and report/notify the relevant parties – this includes the CI of any breaches or incidents related to the study.

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1.1 Abbreviations and acronyms

aDSM	Active TB drug safety monitoring and management
ADR	Adverse drug reaction
AE	Adverse Event
AMR	Antimicrobial resistance
ART	Anti-retroviral therapy
ARV	Anti-retroviral
BMI	Body Mass Index
BPNS	Brief peripheral neuropathy screen
BTS MDR-TB CAS	British Thoracic Society MDR-TB Clinical Advice Service
CTCAE	Common Terminology Criteria for Adverse Events
DAIDS	Division of AIDS
DMID	Division of Microbiology and Infectious Diseases
DR-TB	Drug-resistant Tuberculosis
DST	Drug Susceptibility Testing
ECG	Electrocardiogram
EMA	European Medicines Agency
FDA	United States Food and Drug Administration
GDI	Global Drug-Resistant TB Initiative – Stop TB Partnership
GFATM	Global Fund to Fight AIDS, TB and Malaria
HCW	Health care worker
HIV	Human Immunodeficiency Virus
HRQoL	Health-related quality of life
HR-TB	Isoniazid-mono-resistant tuberculosis
KNCV	KNCV Tuberculosis Foundation
LSHTM	London School of Hygiene & Tropical Medicine
MIC	Minimum inhibitory concentration
MSF	Médecins Sans Frontières
MDR	Multidrug-resistance
MDR-TB	Multidrug-resistant tuberculosis
MTB	Mycobacterium tuberculosis
NHS	National Health Service
QTcF	Heart-rate corrected QT interval
RR-TB	Rifampicin-resistant tuberculosis
SAE	Serious Adverse Event
SOC	Standard of care
SRF	Social risk factor
STR	Shorter treatment regimen
TB	Tuberculosis
TSH	Thyroid Stimulating Hormone
UKHSA	United Kingdom Health Security Agency
USAID	United States Agency for International Development
WHO	World Health Organization
WP	Work package
XDR	Extensive Drug Resistance
XDR-TB	Extensively Drug-resistant Tuberculosis

1.2 Abbreviations of TB agents

Am	Amikacin
Bdq	Bedaquiline
Cs	Cycloserine
Cm	Capreomycin
Cfz	Clofazimine
Dlm	Delamanid
E	Ethambutol
Eto	Ethionamide
FQ	Fluoroquinolone
INH	Isoniazid
Km	Kanamycin
Lfx	Levofloxacin
Lzd	Linezolid
Mfx	Moxifloxacin
Ofx	Ofloxacin
Pa	Pretomanid
PAS	Paraaminosalicylic acid
Pto	Prothionamide
R,RIF	Rifampicin
S	Streptomycin
Tzd	Terizidone
Z	Pyrazinamide

1.3 Protocol at a glance

This document describes a protocol for the conduct of operational research to assess the effectiveness, safety, acceptability, impact on quality of life and cost-effectiveness of treatment for patients with drug-resistant tuberculosis and for patients with drug-susceptible tuberculosis who require second-line medications due to drug intolerance.

Part A of this document describes Work Package (WP) 1, which will investigate the effectiveness and safety of different second-line TB regimens.

Part B of this document describes three complementary work packages evaluating: Health-related quality of life (WP 2a), Stigma (WP 2b), Acceptability to healthcare providers (WP 3) and Cost-effectiveness (WP 4).

TITLE	TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom.
INVESTIGATOR/STUDY LOCATION	<i>England, Northern Ireland, Wales and Scotland</i> Study sites (<i>all to be confirmed</i>): • University College London NHS Foundation Trust • Barts Health NHS Trust • Cambridge University Hospitals NHS Foundation Trust • Cardiff and Vale UHB • Chelsea & Westminster Hospital NHS Foundation Trust • Hull University Teaching Hospitals NHS Trust • Imperial College Healthcare NHS Trust • Leeds Teaching Hospitals NHS Trust • London North West University Healthcare NHS Trust • Ninewells Hospital, NHS Tayside • Royal Free London NHS Foundation Trust • Sheffield Teaching Hospitals NHS Foundation Trust • St George's University Hospitals NHS Foundation Trust • University Hospitals Birmingham NHS Foundation Trust • University Hospitals Bristol and Western NHS Foundation Trust • Whittington Health NHS Trust
STUDY RATIONALE	New all-oral shorter treatment regimens for rifampicin-resistant (RR) and multidrug-resistant (MDR) tuberculosis (TB) are increasingly being used by national TB programmes under operational research conditions. This protocol aims to provide a standardised methodology for conducting such operational research so that the data generated are harmonised both across the UK and globally. The harmonised data will complement evidence from recent clinical trials in high-burden, low-resource settings, to support and inform future guideline development in the UK and globally. For patients requiring second-line regimens due to drug resistance or intolerance, practice across the UK is varied, reflecting persisting uncertainties about the optimum management. Robust locally derived operational data will help to inform national guidelines and harmonise care across the UK.
STUDY OBJECTIVES	For novel, short-course, all-oral DR-TB regimens in the UK, this study aims: 1. to determine whether the treatment effectiveness falls within the range demonstrated in studies in high-burden settings; 2. to define their tolerability, including the burden of adverse effects, the impact upon treatment adherence and requirement for regimen switching as compared to the previous standard of care; 3. to assess the key patient-centred outcomes of health-related quality of life and stigma, using both established and novel tools; 4. to explore healthcare providers' perspectives on the reasons for low uptake, in order to develop strategies to address these barriers;

	5. to evaluate the incremental cost-effectiveness as compared with conventional longer regimens
STUDY DESIGN	Prospective, observational cohort study evaluating all non-first-line treatment regimens in use by participating TB centres in the UK.
STUDY POPULATION	The study population includes all patients receiving treatment for tuberculosis with a second-line regimen due to either: i) resistance to at least one first-line medication (R, INH, Z or E; confirmed by conventional drug susceptibility testing (DST) or rapid molecular DST); or ii) intolerance to rifampicin necessitating the use of second-line medications.
Main inclusion criteria	All participants must be: 1. Willing and able to give informed consent to be enrolled in the research project and follow-up (signed or witnessed consent if the patient is illiterate; signed or witnessed consent from a child's parent or legal guardian); 2. On treatment for tuberculosis; and 3. Unable to take a first-line regimen due to drug resistance or intolerance to first-line medications.
Main exclusion criteria	There are no exclusion criteria.
Total expected number of patients	Approximately 460 patients over the course of 2 years would be eligible for enrolment (100 MDR/RR-TB; 300 INH-mono-resistant TB; 60 rifampicin-intolerant)
Expected number of study sites	20 NHS sites have agreed in principle to participate.

Treatment regimens	This study is observational and participants will continue to be treated as per their current local practice. Participation in the study does not mandate regimen choice. As per NICE guidelines (2016), all patients with MDR-TB should be discussed “at the regional multidisciplinary TB team meeting”. Subject to input from the British Thoracic Society (BTS) MDR-TB Clinical Advice Service, possible starting regimens might include:		
	Possible starting regimens		Indications[†]
	6-month regimen A	6 months Bedaquiline + Pretomanid + Linezolid + Fluoroquinolone	RR-TB, MDR-TB, Drug-intolerance

	6-month regimen B	6 months Bedaquiline + Pretomanid + Linezolid	Pre-XDR-TB, XDR-TB, Drug-intolerance
	9-month regimen	4-6 months Bedaquiline + Fluoroquinolone + Linezolid + Clofazimine + Pyrazinamide + Ethambutol + high-dose Isoniazid; Followed by 5 months Fluoroquinolone + Clofazimine + Pyrazinamide + Ethambutol	RR-TB, MDR-TB
	Long-course regimen	5 active drugs for 6 months; followed by 3 active drugs for 12-18 months Drugs chosen from Groups A, B, C in preferential order and according to susceptibility testing [†]	RR-TB, MDR-TB, pre-XDR-TB, XDR-TB, Drug-intolerance
	HR-TB regimen A	6 months Fluoroquinolone + Rifampicin + Ethambutol + Pyrazinamide	HR-TB
	HR-TB regimen B	9 months Rifampicin + Pyrazinamide + Ethambutol	HR-TB
	[†] HR-TB = isoniazid-resistant, rifampicin-susceptible, fluoroquinolone-susceptible; RR-TB = rifampicin-resistant, isoniazid-susceptible, fluoroquinolone-susceptible; MDR-TB = rifampicin-resistant, isoniazid-resistant, fluoroquinolone-susceptible; pre-XDR / XDR-TB = rifampicin-resistant, isoniazid-resistant, fluoroquinolone-resistant +/- injectable-resistant; Drug-intolerance = drug-susceptible TB but unable to take first-line regimen due to drug intolerance. [‡] Group A: Bedaquiline, Fluoroquinolone, Linezolid; Group B: Clofazimine, Cycloserine; Group C: others.		
EVALUATION CRITERIA	Primary outcomes Treatment regimens will be compared for: <i>i)</i> Effectiveness: the proportion of patients who have a favourable treatment outcome, defined as “cured” or “treatment completed” without recurrence up to 12 months after the end of the treatment. <i>ii)</i> Safety: the proportion of patients who have a serious adverse event up to 6 months after the end of the treatment. <i>iii)</i> Health-related quality of life and stigma, evaluated using existing assessment tools <i>iv)</i> Healthcare worker perspectives on treatment regimens and barriers to adoption assessed through qualitative study using surveys and semi-structured interviews <i>v)</i> Cost-effectiveness in which the incremental cost-effectiveness of novel, shorter regimens will be compared with conventional longer regimens		
ASSESSMENT SCHEDULE	See timeline below		
STATISTICAL CONSIDERATIONS			

Sample size	There is no <i>a priori</i> sample size calculation. Sample size will be dictated by the incidence of drug-resistant TB in our centres and study enrolment rates. However, assuming similar clinical effectiveness to that seen in trial settings, a cohort of at least 62 patients on short-course regimens would be required to identify a point-estimate for effectiveness that is within a 7.5% margin of the 90% efficacy seen in trial conditions, with 95% confidence. For the incidence of Grade 3 or higher adverse events, a sample size of 128 patients would be needed to identify a point-estimate within a 7.5% margin of the 25% incidence seen in the Zenix-TB study, with 95% confidence
Sampling/recruitment strategy	All patients who meet the eligibility criteria will be invited to participate in this operational research study.
DURATION OF STUDY	✓ 6 to 24 months of treatment (based on regimen chosen by treating team) ✓ Up to 12 months of follow-up after the end of the treatment
ANTICIPATED STUDY DATES	Start date: September 2024 Enrolment end date: September 2026 Study end date: December 2028

1.4 Overview of the timelines of the different study components (Part A and Part B)

PART A of the protocol describes work package 1 (WP 1) which is the evaluation of regimen safety and effectiveness. PART B of the protocol describes WP 2 (health-related quality of life and stigma), WP 3 (acceptability to healthcare workers) and WP 4 (cost-effectiveness).

The timelines presented in this table are for both Part A and Part B of the protocol.

The duration of the treatment phase will be at the discretion of the treating centre depending on their chosen regimen.

	Enrolment	Treatment Phase									Follow-up	
Study component		Mt-1	Mt-2	Mt-3	Mt-4	Mt-5	Mt-6	Mt-7	Mt-8	Mt-9 to 24 [†]	Mf-6	Mf-12
Safety	WP1											
Effectiveness ^F	WP1											
Stigma [§]	WP2											
HRQoL	WP2											
Acceptability [§]	WP3											
Cost-effectiveness: Provider	WP4											

Mt-*n* = *n* months into treatment

Mf-*n* = *n* months into follow-up

[†] Where treatment regimens extend beyond 9 months (Mt-9), safety and effectiveness data will continue to be collected monthly. Stigma and health-related quality of life (HRQoL) data will be collected at Mt-12, Mt-18 and Mt-24 where applicable. 'Cost-effectiveness: Provider' will be collected at the end of treatment.

^F The proposed follow-up is 2-4 visits within one year.

[§] This work package relates to healthcare workers. Eligible healthcare workers will be invited to an initial interview and a subsequent interview 12-18 months later.

1.5 Public / Patient Involvement in the Protocol Development

An initial patient consultation has helped to inform the adaptation of this protocol. Five current and former patients with drug-resistant tuberculosis were consulted to discuss the study plan and its conduct. The accompanying patient information sheet and informed consent forms have also been reviewed and amended in line with feedback from our public contributors.

Public and patient involvement (PPI) will continue throughout the study as outlined in Section 16.2.

PROTOCOL PART A: Work Package 1 – Evaluating the safety and effectiveness of differing regimens for drug-resistant tuberculosis in the United Kingdom

2. Schedule of recommended examinations during treatment and follow-up phases of the study

	Investigation/Observation	Baseline assessment & Screening	Treatment Phase (M=Month)									Follow-Up	
			Mt-1	Mt-2	Mt-3	Mt-4	Mt-5	Mt-6	Mt-7	Mt-8	Mt-9 to 24	Mf-6	Mf-12
Clinical evaluation	Demographics, Medical History	X											
	Clinical Examination*	X	X	X	X	X	X	X	X	X	X	X€	X€
	Written informed consent	X											
	Treatment adherence		X	X	X	X	X	X	X	X	X		
	Concomitant treatment		X	X	X	X	X	X	X	X	X	X	
	Adverse events		X	X	X	X	X	X	X	X	X	X	
Bacteriology	Sputum smear	X (2)	X	X	X	X (2)	X	X	X	X	X (2)	X	X
	Sputum culture	X	X	X	X	X	X	X	X	X	X	X	X
	DST (FQ/Injectables)	X										X±	X±
Laboratory tests	Haemoglobin / platelets count / white blood count	X	X#	X#	X#	X#	X#	X#	X#	X#	X#		
	Serum creatinine (at baseline and if clinically indicated or ECG abnormalities)	X											
	Serum potassium (at baseline and if clinically indicated or ECG abnormalities)	X											
	Serum liver enzymes	X	X	X	X	X	X	X	X	X	X		
	Pregnancy test (female) ^ψ	X											

	HIV and hepatitis test	X μ											
	TSH (Note: for patients receiving Pto/Eto)	X			X			X					
Other	Chest X-ray ξ	X											
	ECG	X	X β	X	X	X	X	X	X	X	X		
	Visual acuity & BPNS* (Note: for patients receiving Lzd and high-dose INH/EMB)	X	(X)	(X)	(X)	(X)	(X)	(X)	(X)	(X)	(X)		
	Audiometry for patients receiving injectables (At baseline and monthly afterwards until end of treatment with injectable agents)	(X)	(X)	(X)	(X)	(X)	(X)	(X)	(X)	(X)	(X)		
	Disability assessment	X									X	X	X

* Patients will be examined at least once a week for the first month of treatment and thereafter monthly throughout treatment. This task can be shifted to any health care staff trained and supported to interview and conduct basic examination to detect adverse events. For patient taking linezolid or ethambutol, peripheral neuropathy screen and visual acuity and colour-blindness screening should be performed every month until linezolid and ethambutol are stopped.

Ψ A pregnancy test may also be done at a later stage during the course of treatment as deemed appropriate by the treating clinician.

€ Telephone contact might be accepted if patient cannot attend the TB centre.

$\pm$ DST performed at baseline (with storage of the strain at -20c), after any culture reversion during treatment, and if any culture is found positive during the 12 months' period of follow-up after the end of the treatment (with storage of the strain of the recurrent episode for genotyping analysis).

To be performed if the patient is taking linezolid.

μ Hepatitis B (HepBsAg) and hepatitis C (HCV Ab) tests.

B Baseline ECG should be obtained and additional ECGs conducted at week 1 and 2 after starting treatment and thereafter monthly throughout treatment. ECG should be repeated as necessary in case of clinical suspicion of heart rhythm and conduction disturbances, or other clinical signs (e.g. dehydration and electrolyte misbalance). If delamanid is administered sequentially after treatment with bedaquiline, then an enhanced ECG monitoring should be implemented as bedaquiline has a very long (five month) tissue half-life. Weekly for 1 month then monthly thereafter.

F The optional proposed follow-up is 4 visits every 3 months for one year (or 2 visits every 6 months depending on current practice).

Ξ Chest X-ray may be repeated during follow-up in case of suspicion of recurrence.

Note: Please note that while all the information collected during screening is valuable for subgroup analysis, virtually none is a pre-requisite to ongoing involvement in the study. It is desirable to conduct these examinations before treatment is initiated, but treatment initiation should NOT be delayed in order to complete these.

3. Background

3.1 Drug-resistant tuberculosis

M. tuberculosis (TB) kills more people globally than any other pathogen and drug-resistant tuberculosis (DR-TB) accounts for one-third of antimicrobial resistance (AMR) deaths. DR-TB is a public health crisis and a global health security risk carrying grave consequences for those affected. Globally, 186 772 cases of multi-drug (MDR) or rifampicin-resistant (RR)-TB were detected and notified in 2018, of which 97% were enrolled on treatment with a second-line regimen. This accounts for only a third of the estimated 484 000 people who developed MDR/RR-TB in 2018. An estimated 214 000 people died as a result of it.

In England, isoniazid-resistant TB accounts for 6.4% of cases and MDR/RR-TB for 2.4% of cases (data from UKHSA).¹ A further cohort of patients with drug-susceptible TB also require treatment with second-line regimens due to intolerance to rifampicin.

MDR/RR-TB cannot be treated with the standard 6-month course of first-line medication which is effective in most TB patients. Patients with MDR/RR-TB are treated with a different combination of drugs, which usually has an intensive phase of treatment of 8 months and a total duration of treatment of 20 months (this is referred to as the WHO 2011 long regimen). Outcomes with this approach are generally poor, with only 60% of MDR/RR-TB patients reported to have been successfully treated in the most recent WHO Global TB Report.²

Attempts to reduce the length of conventional MDR/RR-TB regimens and to use a combination of drugs which is tolerable have been ongoing for several years through various studies.

3.2 The WHO consolidated guidelines on DR-TB treatment

In March 2019, WHO released updated, more detailed guidelines on treatment of MDR/RR-TB. These improved guidelines are expected to lead to major improvements in treatment outcomes (given shorter treatment duration and reduced toxicity), and quality of life of MDR/RR-TB patients, including reduced socioeconomic impact. The new guidance announced a priority ranking for medicines available for treatment. Bedaquiline has been recommended to be used as a core drug in the longer TB treatment regimen for MDR/RR-TB patient. It should be prioritized when constructing a longer regimen. Linezolid has been also recommended to be used as a core drug in the longer TB treatment regimen for MDR/RR-TB patients. Kanamycin and Capreomycin should no longer be used in treatment of MDR/RR-TB. Thus, the proposed new treatment regimen includes either an oral medicine only regimen that is longer than 18 months or a 9-month regimen that contains a second line injectable (Amikacin). Neither option is optimal.

New guidelines were published in June 2020: the WHO Consolidated Guidelines on Tuberculosis, Module 4: Treatment – Drug-Resistant Tuberculosis Treatment, and were subsequently updated by a rapid communication in May 2022.^{3,4} These documents include two new recommendations, one on the composition of shorter regimens and one on the use of the BpaL(M) regimen (i.e. bedaquiline, pretomanid and linezolid with or without moxifloxacin).

3.3 Operational research on the all-oral shorter MDR/RR-TB treatment regimens

Some of the regimens currently under trial do not include the use of a second-line injectable, and can potentially be programmatically easier to administer, while at the same time can improve treatment outcomes and quality of life of patients. Although these regimens are currently undergoing testing in clinical trials, the WHO is recommending their use under operational research conditions.⁴ Evidence from this research can inform

programmatic implementation in the United Kingdom and provide important data to the global TB community to strengthen the evidence base and inform treatment guidance. Some of these regimens are already in use in the UK, for example the WHO recommended 9-12 month regimen, but there is not yet a nationwide programme to monitor these.

3.4 Rationale for research on the all oral shorter MDR/RR-TB treatment regimen in the United Kingdom

It is reasonable to assume that individuals treated for MDR/RR-TB with all-oral shorter regimens would experience better quality of life compared to patients on a standard (either short or long) MDR/RR-TB regimen, through two primary mechanisms. First, an injectable-free treatment itself can potentially improve treatment adherence of patients and facilitate the implementation of community-based models of care. This in turn would reduce direct costs incurred by the patients associated with travelling to the health facility for daily injections (such as transportation, food, and accommodation costs), and/or the opportunity costs associated with lost productivity and income due to hospitalization in the intensive phase of treatment.

Second, injectable-free treatment regimens that potentially reduce occurrence of adverse reactions would likely improve adherence to treatment. It is therefore likely that less QALY would be lost and overall health-related quality of life of patients would improve. These hypotheses need to be corroborated by further evidence to inform national policy. Finally, the effectiveness of shorter MDR/RR-TB treatment regimens needs to be ascertained under programmatic and well-resourced conditions.

3.5 TB and DR-TB epidemiology in the United Kingdom

TB Epidemiology:

In 2018, England had the highest rate of tuberculosis in the UK, with a rate of 8.3 per 100,000 population, followed by Scotland (4.9 per 100,000 population), Wales (3.1 per 100,000 population) and Northern Ireland (3.0 per 100,000 population).⁵

In England there has been a steady reduction in TB incidence following a peak in 2011 of 15.6 per 100,000 population (8,280 notifications). Apart from a brief increase in 2019, there was a steady year-on-year reduction to an incidence of 7.3 per 100,000 in 2020 (4,125 notifications).¹ By the end of June 2022, there were 2,226 reports of TB in England.⁶ This puts England in the WHO category of a low incidence country.

In 2020, 59% of patients diagnosed with TB were male. The incidence among people born outside of the UK was 36.3 per 100,000, which is 15-fold higher than among UK-born individuals and accounted for 72.8% of all cases. This pattern differs in children, with two-thirds of paediatric cases reported in children born in the UK, likely reflecting transmission within the UK.¹

Among people born in the UK, the highest incidence was seen in those aged 15 to 34 and those aged 75 years and older (3.0 per 100,000 in both age groups). For those born outside the UK, the highest incidence was in those aged 20 to 24 years (48.4 per 100,000).¹

Of all people with TB, 32.3% (1,332 out of 4,125) were known to have at least one co-morbidity, the most frequent of which was diabetes (11.9%, 466 out of 3,920). Up to 12% of patients diagnosed with TB aged > 15 years in England have at least one social risk factor, including drug misuse, alcohol misuse, homelessness and imprisonment.¹

In 2021, 62.5% of people (2,672 out of 4,285) notified with TB had their diagnosis confirmed by culture³. Culture confirmation is generally higher among people with pulmonary TB compared to those with extra-pulmonary TB

(75.3% vs. 44.2% in 2020). This trend was reversed in children where the proportion of culture confirmation was low for both pulmonary and extra-pulmonary TB (27.1% and 32.7% respectively in 2020).¹

MDR/RR-TB Epidemiology:

In England, preliminary data from 2021 show that 53/2676 (1.9%) people with TB were notified with MDR/RR-TB, compared to 58/2450 (2.4%) notifications in 2020.¹ As of 30 June 2022, 17 cases of MDR/RR-TB have been notified in England.⁶ MDR/RR-TB represented 1.1% of isolates in Scotland 2019;⁷ 2% in Northern Ireland in 2017 and 3% in Wales in 2018 (RR-TB).^{5,8}

The incidence of MDR/RR-TB was higher among those with a previous history of TB compared to those being diagnosed for the first time (5.5%, 7 out of 128 versus 2.2%, 48 out of 2199, respectively). Between 2000 and 2020, 0.2% (168 out of 82,061) of people with culture confirmed TB acquired resistance whilst on treatment in England, of which 29.2% (n=49) acquired resistance to rifampicin and 33.9% (n=57) acquired resistance to isoniazid.¹

Among people with MDR-TB tested for resistance to injectables and/or fluoroquinolones in 2020, 22.0% (9 out of 41) were resistant to at least one injectable agent and 22.0% (9 out of 41) were resistant to fluoroquinolone. The resistance patterns of people with pre-XDR-TB is strongly associated with country of birth; most notably for people of Indian, Lithuanian and Romanian origin.¹

In 2017, the proportion of TB patients with MDR/RR-TB was almost two times higher in people with a social risk factor (SRF) compared to those without (2.7% vs. 1.5%). This difference was not, however, seen in 2020. Nonetheless, in 2020, 12.6% of people diagnosed with any form of TB had a recognised SRF.¹

Isoniazid-resistance epidemiology:

In England, preliminary data from 2021 show that 6.4% of patients (157 out of 2,409) with culture-confirmed TB had resistance to isoniazid without evidence of rifampicin resistance.¹ Resistance to isoniazid was reported at 6.0% in Scotland 2019 and 5% in Wales in 2018.^{5,7}

Rifampicin-intolerance:

The rates of rifampicin intolerance are not well reported in the UK. In one study from New York in 2000, 1.9% (68 out of 3520) of patients with drug-susceptible TB had rifampicin discontinued due to adverse reaction, although the authors felt that discontinuation was not necessary in 57% of those cases.⁹

MDR/RR-TB treatment outcomes:

Of people in the 2018 drug-resistant cohort (n=50), 62% had completed treatment by 24 months after treatment start date; 12.0% were still on treatment; 10.1% were lost to follow-up; and 8% had died. Of the 2017 cohort (n=64), 64% patients had completed treatment by 24 months; 12.5% were still on treatment; 10.9% were lost to follow-up; and 10.9% had died.¹

The time to completion of treatment is known for 82% of patients in the 2017 and 2018 cohorts. For the 2017 cohort, only 1 (2.7%) patient completed treatment in less than 12 months. The majority of patients (21 (55%)) completed in 18-24 months and 5 (13.2%) completed their treatment in more than 24 months. In the 2018 cohort, 2 (7.1%) patients completed treatment in less than 12 months, 18 (63%) in 18-24 months and 2 (7.1%) in more than 24 months.¹

3.6 Drug-resistant TB treatment in The United Kingdom

Drug Susceptibility Testing in the UK:

In the UK, the mycobacterium tuberculosis (MTB) strains from all positive TB cultures are sent to the UKHSA national mycobacterial reference laboratory service for whole genome sequencing (WGS).¹⁰ Phenotypic drug

susceptibility testing is only undertaken if known or putative resistance-conferring mutations are identified by WGS for any first line drugs or if sequencing has been technically unsuccessful for any first line drugs.

UK action plan for management of TB:

The Action Plan for England 2021 to 2026 aims to improve the prevention, detection and control of TB in England. The TB Action Plan will support a year-on-year reduction in TB incidence and in-UK TB transmission and enable the UK to meet its commitment to the World Health Organization (WHO) elimination targets by 2035.¹¹

The TB action plan for England 2021-2026 aims to ensure effective management of cases of MDR-TB by increasing the proportion of cases submitted to the British Thoracic Society MDR-TB Clinical Advice Service (BTS MDR-TB CAS) by 5% per year to 100% in 2024 to 2025.¹¹

The BTS MDR-TB CAS is a UK-wide service that provides advice and support to clinicians who encounter MDR/RR-TB cases. It facilitates expert advice on the treatment and monitoring of cases and provides drug-related adverse effects monitoring and education.¹²

Current treatment policy for MDR/RR-TB in the UK:

The latest NICE guidelines on tuberculosis (2016) do not specify any treatment recommendations for MDR-TB; rather they advise that clinicians “ensure all cases ... are discussed at the regional multidisciplinary TB team meeting”.¹³ This is provided through the British Thoracic Society (BTS) MDR-TB Clinical Advice Service (CAS).

NHS England published a Policy Statement regarding the use of bedaquiline and delamanid in 2019, which refers to the 2018 WHO guidelines for MDR/RR-TB treatment.¹⁴ These guidelines classified anti-TB drugs into three groups (A, B and C). Recommended regimens should be constructed giving preference initially to those medications in Group A (bedaquiline, linezolid and a fluoroquinolone), followed by Group B (cycloserine and clofazimine) and only using medications from Group C where necessary (including injectable agents amongst others). Regimens should contain at least 4-5 active drugs for 6 months, followed by at least 3 active drugs to complete 18-20 months treatment (at least 15-17 months after culture conversion).

The latest WHO consolidated guidelines, which were published in 2020 and subsequently updated by a rapid communication in 2022, additionally recommend several shorter all-oral bedaquiline-containing regimens in patients who have not been exposed to more than 1 month of MDR-TB treatment.^{3,4} These regimens, which have not yet been used widely in the UK, include:

- i. 4-6 Bdq-Lfx-Lzd-E-Z-Cfz-INH (high dose) / 5 Lfx-Cfz-E-Z (provided fluoroquinolone resistance excluded)
- ii. 6 Bdq-Pa-Lzd +/- Mfx

Bedaquiline is available in the UK to patients aged 6 years and over, and delamanid to patients aged 3 years and/or 30kg or more. These drugs are given for the necessary time and sequentially, if required, as determined on a case-by-case basis.¹⁴

In the UK, bedaquiline and delamanid are recommended for the treatment of MDR-TB and XDR-TB subject to the following criteria.¹⁴

- 1.- The case must be discussed and treatment agreed with the MDT of the MDR-TB treatment centre or the regional MDT in conjunction with an MDR-TB treatment centre; paediatric cases must also be discussed, and their treatment agreed by the Paediatric Infectious Diseases Centre.
- 2.- The patient must be managed under directly observed therapy.
- 3.- Bedaquiline or delamanid should be used as part of combination therapy in line with WHO-recommended MDR-TB treatment regimens considering:

- a) *In vitro* resistance to a drug or
- b) Known adverse drug reactions, poor tolerance, or contraindication to any component of the combination regimen.

Safety considerations

The following safety criteria must be adhered to for any regimen containing bedaquiline or delamanid:¹⁴

- 1.- Monitoring QT interval prolongation or development of arrhythmia with pre-treatment ECG to determine baseline QT interval and monitoring at two weeks and then monthly over 6 months; repeat if symptomatic or after the addition of any new medication known to prolong QT.
- 2.- If delamanid is administered sequentially after treatment with bedaquiline, then an enhanced ECG monitoring should be implemented as bedaquiline has a very long (five month) tissue half-life. Weekly for 1 month then monthly thereafter.
- 3.- Caution is required if concurrent administration of drugs recognised to prolong cardiac QT interval (e.g. clofazimine, moxifloxacin, neuroleptics and some anti-emetics); if this is unavoidable monitor ECG after the introduction of the drug and monthly thereafter for the duration of treatment.
- 4.- The baseline and ongoing monitoring recommendations of the TB drug monograph, (<http://www.tbdrugmonographs.co.uk>) must be followed.
- 5.- Provider organisations must register all patients using software such as Blueteq and ensure monitoring arrangements are in place to demonstrate compliance against the criteria as outlined.
- 6.- Special caution should be applied when treating those aged over 65 years to ensure that any current medications or co-morbidities are considered.
- 7.- All clinicians prescribing these drugs should refer to the current summaries of product characteristics (SPCs).

Duration of treatment

NHS England have decided that adult patients may be eligible for treatment with bedaquiline or delamanid for longer than 6 months and/or sequentially if they meet any of the following criteria:¹⁴

- 1.- Laboratory confirmed MDR/XDR-TB with resistance to fluoroquinolones or the injectable drugs (kanamycin, amikacin, capreomycin).
- 2.- Where microbiological evidence is lacking but compelling circumstantial evidence indicates very likely MDR/XDR-TB aetiology (e.g. sputum smear negative active disease in a close contact of a patient with laboratory confirmed MDR/XDR-TB). These should be discussed on a case-by-case basis and in conjunction with the Regional MDT and approved by the regional UKHSA lead.
- 3.- Inability to construct a WHO recommended MDR-TB regimen either through phenotypic or genomic defined resistance pattern or because of intolerance / drug interactions (functional MDR or functional XDR-TB).

Continuation of treatment:

- 1.- Treatment of adults with bedaquiline or delamanid should not continue beyond 24 months.
- 2.- Treatment of children with bedaquiline or delamanid for longer than 6 months and/or sequentially is not recommended.
- 3.- Response to treatment will be monitored by culture and smear conversion and imaging (chest X-ray) and drug sensitivity or molecular sensitivity testing will be used to check for resistance. These results will determine if the patient stays on the current treatment regimen, or a change is required.
- 4.- Bedaquiline is used as a first line drug as part of an MDR TB regimen as recommended by WHO 2018 guidance. Delamanid may be used as a third line drug as expertly advised and in line with WHO guidance.
- 5.- Bedaquiline or delamanid **must not** be added alone to a failing regimen.

- 6.- Bedaquiline or delamanid may be used for a maximum duration of 24 weeks (six months) in the first instance. Treatment must be reviewed at least every six months and any extension(s) (up to six months at a time) must be agreed with the Centre and Regional MDT and submitted with justification through the prior approval system. Reasons for use beyond six months may include delayed microbiological response, weak treatment regimens due to intolerance or drug resistance and/or individual risk factors for poor outcomes.
- 7.- Combined use of bedaquiline with delamanid is not recommended within NHS policy.¹⁴

Treatment stopping criteria:

Treatment will be stopped in the following instances:¹⁴

- 1.- A patient is cured (as defined in the treatment guidance (WHO, 2018))
- 2.- Maximum duration of treatment of up to 24 months is reached.
- 3.- Toxicity or serious side effects caused by bedaquiline or delamanid
- 4.- Sustained QT interval prolongation to > 500ms or the development of a significant arrhythmia.
- 5.- Patients are not complying with directly observed therapy.

Treatment stopped due to serious side effects, toxicity or safety concerns can potentially be reinstated once these have been addressed.¹⁴

Treatment pathways for MDR/RR-TB in the UK:

In the UK, patients with MDR/RR-TB are referred to MDR-TB centres with established experience and expertise in managing and supporting patients with MDR and XDR-TB. The local centres (or MDR-TB treatment centres) provide routine case management, DOT, monitoring of treatment and adverse reactions, patient support and outreach. MDR-TB Centres and Paediatric Infectious Diseases services maintain a register of all patients receiving bedaquiline or delamanid. Centres provide prescription and case review at agreed critical points at least 2 months, 4 months, 6 months, 12 months, 18 months and 24 months.¹⁴

Current treatment policy for isoniazid-monoresistant TB (HR-TB):

There are currently no standardised protocols or pathways for management of isoniazid-monoresistant TB (HR-TB) in the UK. The WHO recommends a regimen including a fluoroquinolone instead of isoniazid: 6 R-Z-E-Lfx.³

3.7 Health financing and social protection in The United Kingdom

All TB care in the UK is provided free-of-charge at the point of care delivery through the National Health Service (NHS). This includes all diagnostic and therapeutic services.

3.8 Drug procurement in The United Kingdom

The NHS has commissioned the use of bedaquiline and delamanid for patients with MDR/RR-TB. NHS England, together with the Department for Health and Social Care, have put into place a process that regularly alerts TB services when TB drug shortages are identified, provides timely information on drug availability and advice on actions to take to reduce the impact of drug shortages. The funding and commissioning of these drugs is managed through the relevant local NHS England Specialised Commissioning Team.¹⁴ A commissioning process for procurement of pretomanid is in progress (as of April 2023).

4. Evidence on the drugs proposed in this study

4.1 New and repurposed drugs

4.1.1 Bedaquiline

Chemical composition and dosing

Bdq fumarate (bedaquiline) is a diarylquinoline anti-mycobacterial drug. It inhibits adenosine triphosphate synthesis, a novel method of action. The drug has a 5.5-month half-life. It is indicated for use against MDR/RR-TB (US FDA, EMA). Because it is highly lipophilic, Bdq requires a loading dose of 400 mg daily for 14 days followed by a maintenance dosing of 200 mg thrice weekly.

In the UK, the recommended doses for children aged 6 or more are:¹⁴

- 1-15kg: Consult with a specialist
- 16-23kg: 200 mg daily for the first 2 weeks, followed by 100 mg three times per week
- 24-30kg: 200 mg daily for the first 2 weeks, followed by 100 mg three times per week
- 31-34kg: 400 mg daily for the first 2 weeks, followed by 200 mg three times per week
- >34kg: 400 mg daily for the first 2 weeks, followed by 200 mg three times per week

Efficacy

Strong bactericidal and sterilizing activity against *M. tuberculosis* organisms have been shown in preclinical laboratory setting as well as in animal experiments. Data gathered from clinical trial and operational studies provided enough evidence to classify this agent in group A during the last revision of the WHO guidelines (2019).¹⁵ There is reported cross-resistance of Bdq with clofazimine (Cfz) due to mutations in Rv0678, a transcriptional repressor of the genes encoding the MmpS5-MmpL5 efflux pump. Bdq shows linear pharmacokinetics and better absorption when the drug is taken with food versus when taken fasting (resulting in approximately a two-fold increase in serum drug levels).¹⁴

Safety and tolerability

The phase IIB trial of Bdq found higher all-cause mortality among persons who received Bdq compared with placebo —although data from other studies suggest that this finding is not being observed in observational cohorts and program conditions. Bdq is associated with QTc prolongation in anywhere from 5-20% of persons who have received the drug. Such QTc prolongation has not yet been associated with an increase in fatal arrhythmias but it does bear careful monitoring. Bdq may also be associated with elevated transaminase levels but that has been reported with multiple drugs used for the treatment of MDR/RR-TB. For patients on Bdq, it is essential that a baseline QTc interval be assessed with regular follow up while on the drug and the continued monitoring of liver function.

Bedaquiline should be used with caution when given together with drugs that inhibit liver enzyme function (e.g. ketoconazole, efavirenz or lopinavir/ritonavir effect on CYP3A4) as this could increase bedaquiline concentration and toxicity. Concurrent administration of bedaquiline with CYP3A4 inducers (such as the rifamycins) is contraindicated in view of its metabolism via this route.{Citation}

Knowledge about the safety of Bdq in pregnancy and while breastfeeding is sparse. The drug has therefore been classified as Category B as animal reproduction studies have failed to demonstrate a risk to the foetus and there are no adequate and well-controlled studies in pregnant women.

4.1.2 Clofazimine

Chemical composition and dosing

Clofazimine (Cfz) is a lipophilic riminophenazine initially synthesized in 1954 and licensed for treatment of leprosy in 1969. Its mechanism of action remains unclear, but existing evidence suggests production of reactive oxygen species in *Mycobacterium tuberculosis* cells. In vivo efficacy is believed to be driven by a combination of both anti-mycobacterial activity and anti-inflammatory property of clofazimine. Using a mouse model of chronic tuberculosis, Grosset *et al.* demonstrated that a clofazimine-containing regimen was significantly more active in achieving culture conversion and preventing relapse compared to its clofazimine-free regimen. Studies demonstrate a prolonged lag time for absorption, high variability in bioavailability and clearance, and a terminal half-life of 70 days. The effect of treatment duration on efficacy and safety is not known.

Efficacy

The MIC of clofazimine against *M. tuberculosis* ranges from 0.06 to 2.0 µg/mL. In one study, the minimum bactericidal concentration against *M. tuberculosis* ranged from 0.12 to 0.48 µg/ml. Potent activity against hypoxic, non-replicating *M. tuberculosis* suggests clofazimine may have potential as a sterilizing drug. Mechanisms for resistance have not been reported. Following oral absorption, clofazimine is distributed extensively in the body and primarily into fatty tissues and the mononuclear phagocyte system. Oral administration of clofazimine 100, 300 and 400mg daily in leprosy patients resulted in average plasma levels of 0.7, 1.0 and 1.41µg/ml, respectively. In the consolidated WHO MDR/RR-TB guidelines this drug was classified in the group B of agents for the treatment of MDR/RRTB.

Safety and tolerability

Patients taking clofazimine may experience dry skin, and clofazimine causes slowly reversible redblack skin discoloration in virtually all patients treated for more than a few months.

Clofazimine may also prolong the QTc interval. Such QTc prolongation has not yet been associated with an increase in fatal arrhythmias but it does bear careful monitoring. Data on breastfeeding and pregnancy are sparse (some reports of normal outcomes, and some of neonatal deaths). Clofazimine is only recommended during pregnancy when benefit outweighs risk. Infants exposed in utero or during breastfeeding may be more deeply pigmented at birth.

4.1.3 Delamanid

Chemical composition, dosing, and duration

Dlm (OPC-67683 or Deltyba™) is a new agent derived from the nitro-dihydro-imidazo[4,5-b]pyridine class of compounds. It inhibits the synthesis of mycolic acid, which is an exclusive component of the external wall of mycobacteria. Peak plasma levels are achieved 4-5 hours after administration. The half-life of Dlm is 38 hours after drug discontinuation. Further details on dosing are given in Annex 3 and 4.

Efficacy

Dlm has also demonstrated in vitro and in vivo activity against both drug susceptible and drug resistant strains of *M. Tuberculosis*, and was granted conditional approval by the EMA in 2014 based on phase I/II data which demonstrated higher rates of culture conversion, faster time to culture conversion, and higher treatment success rates among person who received less than 2 months of Dlm versus those who received 2 months or more. After the phase I/II data, WHO immediately issued guidelines recommending its use when an effective treatment regimen cannot be designed and in patients with high risk of poor outcome. These guidelines have been later updated in 2016 for the use of Dlm in the treatment of MDR/RR-TB in children and adolescents. In the consolidated 2019 WHO guidelines on DR-TB treatment this drug was classified in the group C of agents for the treatment of MDR/RR-TB. Dlm absorption is increased after a standard meal.

Safety and tolerability

Delamanid has an excellent safety profile and is deemed one of the least toxic agents used in the treatment of MDR/RR-TB. Although early studies showed some QTc prolongation with Dlm, the phase III trial found similar rates of QTc prolongation when comparing the Dlm regimens and the placebo regimens. Caution is required in patients with known cardiac risk factors for QT interval prolongation (e.g. known congenital QTc-interval prolongation or any condition known to prolong QTc interval or QTc > 500 ms; history of symptomatic cardiac arrhythmias or clinically relevant bradycardia; any predisposing cardiac conditions for arrhythmia; electrolyte disturbances, and medicinal products known to prolong QTc interval. Increased ECG monitoring frequency for the full delamanid treatment period should be undertaken in patients who start delamanid with serum albumin 2.8g/dl to 3.4 g/dl or fall into this range. Delamanid should not be administered if serum albumin is <28g/L.¹⁴

Drug–drug interaction studies in healthy subjects show no clinically significant interactions when Dlm is co-administered with CYP3A inhibitors such as tenofovir, efavirenz or lopinavir/ritonavir. Concurrent administration of strong CYP3A inducers (e.g. carbamazepine) is contraindicated in view of delamanid’s metabolism via this route

Knowledge about the safety of Dlm in pregnancy and while breastfeeding is sparse.

Dlm has not been used extensively in children. In the UK, the recommended dosing in children (age 3+ years and/or 30kg or more), by weight is:

- 30-49kg: 50 mg twice daily
- 50kg or above: 100 mg twice daily

4.1.4 Levofloxacin

Chemical composition and dosing

Levofloxacin (Lfx) belongs to the second-generation medicine of the fluoroquinolone class. The medicine was repurposed for treatment of MDR/RR-TB and remains one of the pivotal medicines for MDR/RR-TB treatment. Levofloxacin is a concentration dependent medicine and diffuses through the bacterial cell wall, acts by inhibiting DNA gyrase (bacterial topoisomerase II), an enzyme required for DNA replication, RNA transcription, and repair of bacterial DNA. Inhibition of DNA gyrase activity leads to blockage of bacterial cell growth. Single oral doses of levofloxacin 50 to 1000mg produce a mean C_{max} and area under the concentration-time curve (AUC) ranging from approximately 0.6 to 9.4 mg/L and 4.7 to 108 mg.h/L, respectively, both increasing linearly in a dose-proportional fashion. Levofloxacin is widely distributed throughout the body, with a mean volume of distribution of 1.1 L/kg, and has excellent penetration into chronic cavitory TB lesions.

The plasma elimination half-life ranges from 6 to 8 hours in individuals with normal renal function (80% of levofloxacin is eliminated as unchanged drug in the urine through glomerular filtration and tubular secretion). The effect of duration on efficacy and safety is not known.

Efficacy

Levofloxacin has strong anti-TB activity. There might be cross-resistance with other fluoroquinolones but it may not be complete. Data suggests greater activity than ciprofloxacin or ofloxacin. This drug is classified in the group A of agents for the treatment of MDR/RR-TB in the consolidated 2019 WHO MDR/RR-TB guidelines.

Safety and tolerability

Levofloxacin can cause QTc prolongation but less frequently than Moxifloxacin. Such QTc prolongation in fluoroquinolones has been associated with an increase in fatal arrhythmias, and it warrants careful monitoring.

Levofloxacin may potentially interfere with lamivudine clearance (increasing the levels of lamivudine), but is not contraindicated with other antiretroviral agents and no drug dosing adjustments are needed. Co-administration of levofloxacin with oral divalent cation-containing compounds (such as antacids) may impair its absorption and should be avoided. Dosage adjustments are required in patients with significant renal dysfunction.

Fluoroquinolones have been associated with arthropathy in puppy models. However, there are case reports of fluoroquinolones being used safely during pregnancy.

4.1.5 Linezolid

Chemical composition and dosing

Linezolid (Lzd) is an oxazolidinone antibiotic that inhibits bacterial protein synthesis by binding 23S ribosomal RNA, and is active in vitro against *M. tuberculosis*, including MDR and XDR-TB strains, at concentrations of 1 µg/mL or less in most studies. Lzd has been used in combination with other second- and third-line anti-TB drugs in multidrug regimens for the treatment of MDR and XDR-TB with variable success; however, rigorous evaluation of the safest and most effective dose, the dose response relative to TB outcomes, the singular contribution of Lzd when added to other active drugs in a regimen, and the use of Lzd together with other new anti-TB drugs is limited.

Currently it is recommended that the drug be started at a dose of 600 mg daily and given for the entire course of therapy: linezolid can either be decreased to 300 mg daily or stopped if limiting toxicity develops (more details on dosing are included in Annex 2 and 3).

Efficacy

Linezolid is an antibiotic that has been demonstrated in two randomized controlled trials and in observational studies to increase culture conversion and treatment success in DR-TB patients. It is considered to be a highly effective agent, but its use is limited by safety concerns (see below). This drug is classified in the group A of agents for the treatment of MDR/RR-TB in the consolidated 2019 WHO MDR/RR-TB guidelines.

Safety and tolerability

The toxicity profile of linezolid – which includes partially reversible myelosuppression, optic neuritis, neuropathy, and lactic acidosis – limits its use. Studies have shown that treatment limiting toxicity can occur in as many as 11% of persons who receive treatment with linezolid. Adverse events mostly occur when linezolid is given at doses of more than 600 mg a day, but they can usually be identified early with routine monitoring and are often reversible upon discontinuation of the drug or lowering of the dose. Pyridoxine (vitamin B6) could be used when linezolid is administered to minimise side effects. Knowledge about the safety of linezolid during pregnancy and breastfeeding is limited so caution is advised.

4.1.6 Moxifloxacin

Chemical composition, dosing and duration

Fluoroquinolones are potent inhibitors of DNA gyrase and topoisomerase IV and several quinolones have demonstrated activity against *M. tuberculosis*. Moxifloxacin (BAY 12-8039) is an 8-methoxyquinolone that is highly active against Gram-positive and Gram-negative bacteria and anaerobes. It has a long half-life of 11.5-15.6 hours that allows it to be given once a day. The effect of duration on efficacy and safety is not known.

Efficacy

Moxifloxacin has anti-TB activity and achieves high levels in tissues including the lung. Moxifloxacin is available as a 400mg tablet that is rapidly absorbed and is excreted in urine and faeces. Moxifloxacin has contributed to treatment success when used off-label as part of regimens treating M/XDR-TB in humans. Later generation

fluoroquinolones have been demonstrated to play an important role in the clinical management of MDR/RR-TB and have been recommended for inclusion in the development of MDR/RR-TB regimens. This drug is classified in the group A of agents for the treatment of MDR/RR-TB in the consolidated 2019 WHO MDR/RR-TB guidelines.

Safety and tolerability

In clinical trials with various indications of moxifloxacin 400mg daily, the most common adverse drug reactions (>3%) were nausea, diarrhoea, headache and dizziness. Tendinopathy/tendon rupture and QTc prolongation are some of the serious adverse drug reactions that may occur with moxifloxacin. ECG monitoring of the QT interval in patients on long-term moxifloxacin is recommended and the drug should not be used in persons with risk factors predisposing to the development of arrhythmias.

Fluoroquinolones are generally avoided during pregnancy and breastfeeding due to observation of arthropathy in puppy models. However, there are a few case reports of fluoroquinolones being used safely during pregnancy.

4.1.7 Pretomanid

Pretomanid is a nitroimidazole (same chemical class as Delamanid) and it is a prodrug that is metabolically activated by a nitroreductase, producing various metabolites that are responsible for its therapeutic action. Pretomanid inhibits cell wall biosynthesis and under anaerobic conditions, it causes respiratory poisoning of the bacterial cell through the release of reactive nitrogen species. Pretomanid was approved in August 2019 by the US FDA in combination with bedaquiline and linezolid as part of the BpaL regimen (also referred to as Nix-TB), an all-oral 6-9-month regimen for the treatment of XDR-TB or treatment-intolerant/non-responsive MDR/RR-TB patients.

Pretomanid is not yet commissioned for use in England on the NHS; however, an application has been submitted and is pending review (as of August 2022).

4.2 Evidence on the other drugs proposed in this study

4.2.1 Isoniazid

Chemical composition, dosing and duration

Isoniazid is a long-standing component of the standard DS-TB treatment regimen. It is a first-line TB medicine that may be an effective second-line agent in the absence of high-level isoniazid resistance.

Isoniazid is a synthetic derivative of nicotinic acid with anti-mycobacterial properties. Although its mechanism of action is still unclear, isoniazid appears to block the synthesis of mycolic acids, major components of the mycobacterial cell wall. This agent is only active against actively growing mycobacteria because, as a pro-drug, it requires activation in susceptible mycobacterial species. Isoniazid also interferes with mycobacterial metabolism of vitamin B6. Resistance occurs due to decreased bacterial wall penetration.

Safety and tolerability

Isoniazid therapy is often associated with minor, transient and asymptomatic elevations in serum aminotransferase levels but, more importantly, isoniazid is a well-known cause of acute clinically apparent liver injury which can be severe and is sometimes fatal.

Adverse reactions include hepatitis (age-related), peripheral neuropathy, and hypersensitivity reactions. Pyridoxine (vitamin B6) should be used when high-dose isoniazid is administered and in patients with diabetes, uraemia, HIV infection, seizure disorders, alcohol abuse, malnutrition or peripheral neuropathy. Additionally, pregnant and postpartum women and exclusively breastfed infants should receive vitamin B6 while taking isoniazid.

4.2.2 Ethambutol

Chemical composition, dosing and duration

Ethambutol is an antibiotic with bacteriostatic, antimicrobial and antitubercular properties. Ethambutol interferes with the biosynthesis of arabinogalactan, a major polysaccharide of the mycobacterial cell wall. This results in halting bacterial growth.

Ethambutol is a first line but adjunctive anti-tuberculosis medication which is used only in combination with other agents such as isoniazid and rifampin.

Safety and tolerability

Ethambutol therapy has been associated with minor, transient and asymptomatic elevations in serum aminotransferase levels, but is a rare cause of clinically apparent acute liver injury.

Adverse reactions include retrobulbar neuritis (dose-related, exacerbated during renal failure). It is recommended to avoid concurrent administration of ethambutol with aluminium hydroxide containing antacids for at least 4 hours following ethambutol administration. There is no reported cross resistance. Patients should be counselled to report any changes in vision. Baseline and monthly visual acuity and colour discrimination monitoring should be performed. The use of ethambutol is safe in pregnancy, and ethambutol can be used while breastfeeding.

4.2.3 Pyrazinamide

Chemical composition, dosing and duration

Pyrazinamide is a synthetic pyrazinoic acid amide derivative with bactericidal property. Pyrazinamide is particularly active against slowly multiplying intracellular bacilli (unaffected by other drugs) by an unknown mechanism of action. Its bactericidal action is dependent upon the presence of bacterial pyrazinamidase, which removes the amide group to produce active pyrazinoic acid. Pyrazinamide is a first line anti-tuberculosis medication, but is used only in combination with other anti-tuberculosis medications such as isoniazid or rifampicin.

Safety and tolerability

Pyrazinamide is associated with transient and asymptomatic elevations in serum aminotransferase levels and is a well-known cause of clinically apparent, acute liver injury that can be severe and even fatal. Pyrazinamide may be used for drug-resistant TB when the isolate is sensitive to pyrazinamide (no known teratogenicity during pregnancy), and it can be used while breastfeeding.

5. Study objectives (Work Package 1)

5.1 Primary objective

To estimate the treatment **effectiveness** (assessed by a composite outcome (“favourable outcome”, defined as “cure” or “treatment completed” without recurrence up to 12 months after the end of treatment)) and **safety** (defined as the occurrence of serious adverse events) of all non-first-line treatment regimens for tuberculosis under programmatic conditions in the United Kingdom.

5.2 Secondary objectives

1.- To determine the proportion of TB patients treated with a non-first-line treatment regimen under programmatic conditions who:

- A) **Died** while on treatment
- B) had **treatment failure**
- C) Had a **recurrent** episode of TB during the 12-month follow-up period
- D) **Relapsed** during the 12-month follow-up period
- E) Were **lost to follow-up** during treatment
- F) Are “**cured without permanent disability**” (up to one year after the end of the treatment)
- G) Complete at least 90% of doses (*treatment adherence*)
- H) Experience adverse events of special interest (see Annex 8)
- I) Are treated with a short-course regimen (6-9 months)
- J) Develop new resistance to an anti-tuberculous drug whilst on treatment
- K) Demonstrate a change in health-related quality of life (HRQoL) scores over treatment

6. Study design

6.1 One single observational cohort

This study will adopt a longitudinal design with a cohort of TB patients receiving non-first-line treatment regimens, either due to resistance patterns or drug intolerance. Figure 1a provides an overview of the study design.

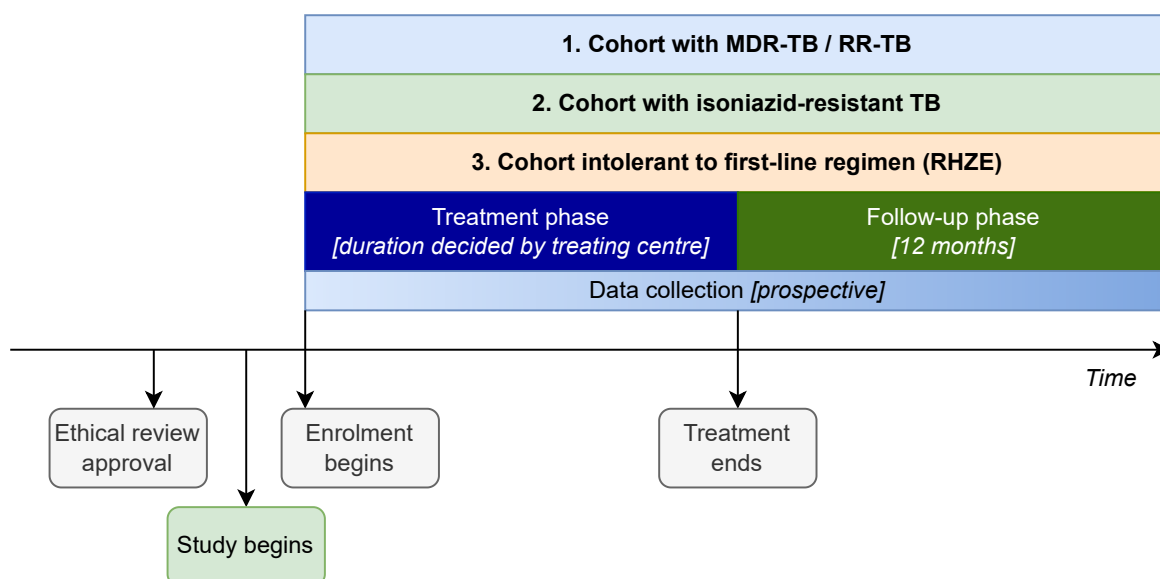


Figure 1a: Overview of the basic study design.

It is anticipated that the treatment regimens in use may evolve during the study period as new drugs, for example pretomanid, become available. In this case, the study will adopt a *de facto* stepped-wedge design, allowing comparison between the older and emerging regimens (see Figure 1.b).

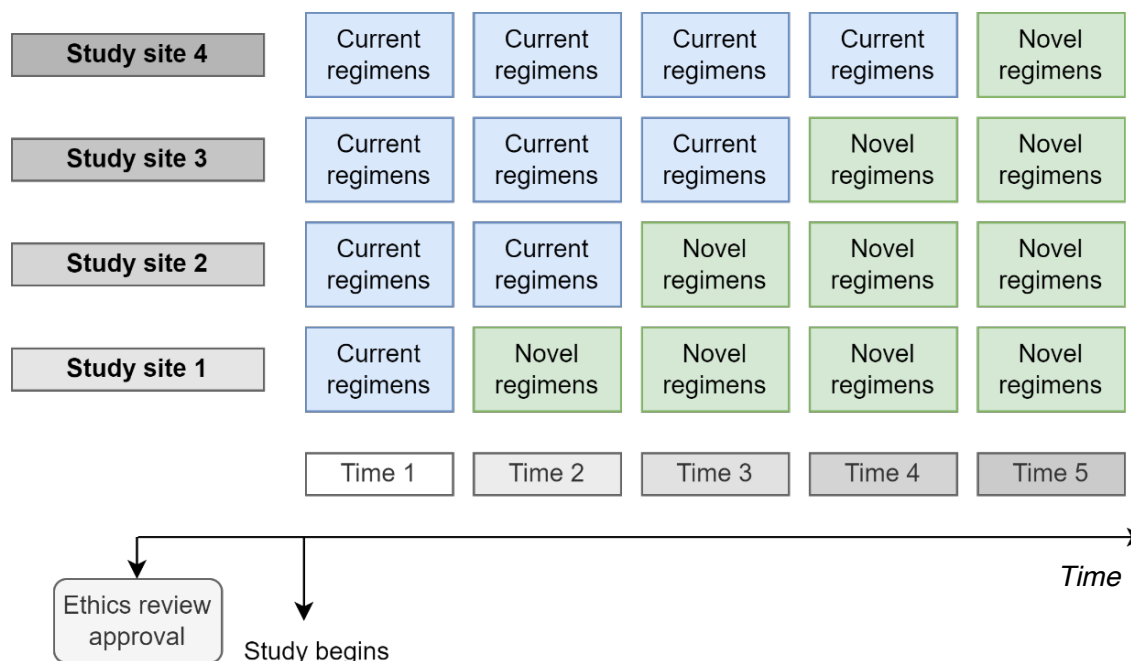


Figure 1b: Stepped-wedge design as emerging treatment regimens become available.

7. Study population

7.1 Study population

The study population includes TB patients with either:

- MDR/RR-TB: evidence of resistance to at least rifampicin by conventional DST (culture-based) or rapid molecular DST; for children, active TB that is likely to be MDR/RR-TB based on history of close contact with a confirmed MDR/RR-TB case; or
- HR-TB: evidence of resistance to isoniazid (but confirmed sensitivity to rifampicin) based on whole-genome sequencing and/or conventional DST (culture-based); or
- Drug-susceptible TB but with intolerance to first-line medications, necessitating use of second-line regimens.

7.2 Inclusion criteria

A patient will be eligible for inclusion in the study, if all the following conditions are satisfied, namely they:

- Are willing and able to give informed consent to be enrolled in the research project and for follow-up (signed or witnessed consent if the patient is illiterate; signed or witnessed consent from a child's parent or legal guardian);
- Are on treatment or about to begin treatment for tuberculosis;
- Require treatment with a non-first-line regimen due to resistance pattern or drug intolerance (as decided by the treating centre).

7.3 Exclusion criteria

There are no exclusion criteria.

7.4 Sample size

All patients who meet the eligibility criteria will be invited to participate in this operational research study.

The number of patients to be invited into this study has been estimated based on surveillance data from the UKHSA 2021 tuberculosis report, specifically the number of bacteriologically confirmed MDR/RR-TB patients, the number of patients with resistance to fluoroquinolones and the number of patients with isolated isoniazid resistance.¹ The proportion of patients who are intolerant to first-line medications is unknown but has been estimated based on an old case series.⁹

Based on these estimates, the number of patients expected to be eligible for this study, per year, will be approximately:

- 50 MDR/RR-TB
- 150 HR-TB
- 30 rifampicin-intolerant

Total: 230 per annum

8. Treatment regimens under investigation

8.1 TB treatment regimens

As this study is observational, the treatment regimens in use will be decided by each centre according to their current practice. As per current practice, it is anticipated that a high proportion of the MDR-TB regimens will be discussed via the BTS MDR-TB CAS.

The Study Management Group has proposed a list of possible starting regimens for patients in the UK, subject to discussion in the national MDR-TB Clinical Advice Service; however, this is not an exhaustive list and practice may vary between centres and between individual patients.

Possible starting regimens		Indications [†]
6-month regimen A	6 months Bedaquiline + Pretomanid + Linezolid + Fluoroquinolone	RR-TB, MDR-TB, Drug-intolerance
6-month regimen B	6 months Bedaquiline + Pretomanid + Linezolid	Pre-XDR-TB, XDR-TB, Drug-intolerance
9-month regimen	4-6 months Bedaquiline + Fluoroquinolone + Linezolid + Clofazimine + Pyrazinamide + Ethambutol + high-dose Isoniazid; Followed by 5 months Fluoroquinolone + Clofazimine + Pyrazinamide + Ethambutol	RR-TB, MDR-TB
Long-course regimen	5 active drugs for 6 months; followed by 3 active drugs for 12-18 months Drugs chosen from Groups A, B, C in preferential order and according to susceptibility testing [‡]	RR-TB, MDR-TB, pre-XDR-TB, XDR-TB, Drug-intolerance
HR-TB regimen A	6 months Fluoroquinolone + Rifampicin + Ethambutol + Pyrazinamide	HR-TB
HR-TB regimen B	9 months Rifampicin + Pyrazinamide + Ethambutol	HR-TB

[†] HR-TB = isoniazid-resistant, rifampicin-susceptible, fluoroquinolone-susceptible; RR-TB = rifampicin-resistant, isoniazid-susceptible, fluoroquinolone-susceptible; MDR-TB = rifampicin-resistant, isoniazid-resistant, fluoroquinolone-susceptible; pre-XDR / XDR-TB = rifampicin-resistant, isoniazid-resistant, fluoroquinolone-resistant +/- injectable-resistant; Drug-intolerance = drug-susceptible TB but unable to take first-line regimen due to drug intolerance.

[‡] Group A: Bedaquiline, Fluoroquinolone, Linezolid; Group B: Clofazimine, Cycloserine; Group C: others.

8.2 Dosing

Annex 2 and 3 describe the drugs included in the proposed regimens, and the duration and dosing of each drug based on the 2019 WHO Consolidated guidelines. Annex 4 presents the provisional, suggested doses of delamanid and bedaquiline for operational research for paediatric patients. The provisional suggested doses of delamanid and bedaquiline are based on the IMPAACT Network trial (for bedaquiline Study P1108).

8.3 Access to the paediatric formulation of delamanid for paediatric patients

Not applicable.

8.4 Procedure following missed treatment

Any missed days will be made up by extending the regimen by the number of days missed but not exceed 10% of the planned study regimen duration. Patients with treatment interruption for two consecutive months or more will be classified as outcome “lost to follow-up”, in which case the patient will be withdrawn from the study and managed on a case-by-case basis, following expert advice, including BTS MDR-TB CAS consultation.

Reasons for missing treatment will be recorded when known, and patients will be included in the acceptability part of the qualitative analysis of this study (see WP3 in part B, section 4).

8.5 Procedure in case of treatment failure

In case treatment failed as defined in Table 1, the status of such patients will be classified as outcome “treatment failure” in the study records. These patients will be managed according to national guidance for the treatment of drug-resistant TB patients, including BTS MDR-TB CAS advice.

9. Outcomes of interest measurement

9.1 Primary outcomes of interest

The following are the two main outcomes of interest of this study:

1. Treatment *effectiveness*: the proportion of patients who have a **favourable treatment outcome**. This is defined as “cured” or “treatment completed” without recurrence up to 12 months after successful treatment (see Table 1).
2. Treatment *safety*: the proportion of patients included in the study with **serious adverse events** occurring during treatment and up to 6 months after the end of the treatment.

Note: Six months is the proposed follow-up time for adverse events due to the half-life of bedaquiline.

9.2 Secondary outcomes of interest

Secondary outcomes of interest of this study include:

1. The proportion of patients who **died** while on treatment.
2. The proportion of patients who had a **treatment failure**.

3. The proportion of patients who had a **recurrent** episode TB during the 12-month follow-up.
4. The proportion of patients who **relapsed** during the 12-month follow-up.
5. The proportion of patients who are “**cured without permanent disability**” (up to one year after the end of the treatment).
6. The proportion of patients who complete at least 90% of doses (*treatment adherence*).
7. The average number of adverse events of interest experienced by patients (see list of adverse events of interest in Annex 8).
8. The proportion of patients experiencing each adverse event of interest (see list of adverse events of interest in Annex 8).
9. The proportion of patients who experience serious adverse drug reactions.
10. The proportion of patients treated with a short-course regimen (6-9 months).
11. The proportion of patients who develop new resistance to an anti-tuberculous drug whilst on treatment.
12. Median change in HRQoL score from beginning of treatment to end of treatment.

10. Key definitions

Table 1 provides the key definitions employed in the study.

Table 1: Key definitions

EVENT	DEFINITION
Favourable outcome	Composite outcome corresponding to the combination of “cured” + “treatment completed” without recurrence during the follow-up period (= treatment success). Note: this outcome can also be defined as “recurrence-free cure”.
Cured	A patient with bacteriologically confirmed TB who has completed a full course of treatment (as determined by the treating centre) without evidence of failure AND at least two consecutive cultures taken at least 30 days apart are negative at the end of the treatment and at least one month earlier.
Treatment Completed	A patient who completes a full course of treatment (as determined by the treating centre) without evidence of failure BUT without bacteriological evidence of cure (i.e. without a negative culture at the end of the treatment phase and at least one month earlier).
Treatment Failed	Treatment terminated or need for permanent change of the regimen protocol of at least two anti-TB drugs because of: • Lack of sputum culture conversion after 4 months of treatment, or • Bacteriological reversion of sputum culture after 5 months of treatment in a patient with previous culture conversion to negative, or • Evidence of additional acquired resistance to drugs in the study, or • Adverse drug reactions (ADRs) (leading to the permanent change of at least two anti-TB drugs in the regimen)
Died	A patient who dies for any reason during the course of treatment.
Lost to follow-up	A patient whose treatment was interrupted for 2 consecutive months or more.
Not evaluated	A patient for whom no treatment outcome is assigned (this includes cases “transferred out” to another treatment unit and whose treatment outcome is unknown/cannot be assessed).
Withdrawn	A patient whose treatment regimen is changed for any reason other than treatment failure (for example, withdrawn patient informed consent or other reasons).

Treatment Success	The sum of <i>cured</i> and <i>treatment completed</i> .
Recurrence	Cure or treatment completion followed by two consecutive positive cultures during post-treatment follow-up (without genotyping information on baseline and recurrent strain), or one positive culture with clinical signs and symptoms or radiographic deterioration.
Relapse	Recurrence in which isolates of the recurrent episode share the same genotype pattern with isolates of the first episode of TB.
Reinfection	Recurrence in which isolates of the recurrent episode and isolates of the first episode of TB have different genotype patterns.
Conversion (to negative)	Culture is considered to have converted to negative when two consecutive cultures taken at least 30 days apart are found to be negative. In such case, the specimen collection date of the first negative culture is used as the date of conversion. In case patients were culture negative at baseline, a negative culture result at month 4 may be considered as “initial conversion”.
Reversion (to positive)	Culture is considered to have reverted to positive when after an initial conversion, two consecutive cultures taken at least 30 days apart are found to be positive. In case of patients who are culture negative at baseline, a positive culture result at month 4 may be considered as “initial reversion”.
Treatment adherence	90% of the treatment doses were taken based on information in the treatment cards or clinical records, measured over the entire treatment period.
Permanent disability	A combined outcome, using the modified Medical Research Council Dyspnoea scale (mMRC) (Annex 10), based on which patients with a score above 2 are considered permanently disabled in terms of their pneumological function. In addition, all serious adverse events by system organ class that are not resolved at the end of follow-up, should be summarised by treatment regimen. This is a measure of a programme’s ability to start treatment promptly and treat patients effectively.
Serious Adverse Event (SAE)	Any untoward medical occurrence that may present in a TB patient during treatment with a pharmaceutical product, but which does not necessarily have a causal relationship with this treatment, which either leads to: • Death • Life-threatening experience • Hospitalization or prolongation of hospitalization • Persistent or significant disability • A congenital anomaly

11. Investigational plan

11.1 Patient enrolment and timeline

Patient enrolment will be ongoing for at least 24 months to gather sufficient information to draw conclusions. Given that most patients will complete treatment within 9-12 months and enrolment will continue for 24 months, it is expected that most data collection and primary data analysis could be completed 36 months after the start of the study.

Patients who meet the inclusion criteria and give consent will be enrolled in the study as they present at the health centre for treatment during the study period.

All drug-resistant TB patients presenting at participating treatment sites will be referred for the screening process. In addition, patients with drug-susceptible TB who develop intolerance to first-line medications during the course of their treatment will also be referred for screening at the time of regimen change.

Once the patient has been found eligible, he/she will be offered participation in the study. Written, informed consent will be obtained before any data collection is carried out. Each participant will receive treatment according to the usual practice of their treating centre and they all will be followed for 12 months after the end of treatment.

Patients who decline participation in the study or are not eligible will continue with their routine treatment and care with no negative consequences for the patient. For all patients, routine procedures in doing contact tracing among household members will be done.

The study timetable at the beginning of this protocol (Section 2.1) details the investigations and observations to conduct at each visit.

11.2 Screening and examinations at baseline

Consent from the patient (or from the guardian for children) will be sought for all patients fulfilling inclusion criteria. Clinical evaluation, bacteriological and other laboratory tests, a chest X-Ray (or two if chest X-Ray is repeated during follow-up in case of suspicion of recurrence), and a 12 lead ECG will be performed. These are detailed below.

- a. **Demographic characteristics and medical history** will be recorded.
- b. **Clinical evaluation.** This will include:
 - Height and weight
 - Vital signs (blood pressure, temperature)
 - Clinical examination (particularly Brief peripheral neuropathy screen, BPNS)
- c. **Bacteriological tests**, including:
 - Sputum smear (x 2)
 - Sputum culture
 - Freeze baseline culture
 - DST (Xpert MTB/RIF, LPA, culture-based first and second line DST)
- d. **Laboratory testing**, including:
 - Full blood count (Haemoglobin/platelets count / White blood count);
 - Serum creatinine
 - Biochemistry (Serum potassium)
 - Serum liver enzymes (ASAT, ALAT)
 - Thyroid stimulating hormone (*if Pto/Eto included in regimen*)
 - HIV test (and CD4 count and viral load, if HIV-positive and not recently done)
 - Hepatitis test (HBs Ag and HepC Ab)
 - Pregnancy test
- e. **Other Exploratory tests:**
 - Chest X-ray

- ECG
- Visual acuity and colour-blindness screen
- Audiometry (if injectable agents are part of the treatment regimen)

11.3 Examinations during treatment

All patients enrolled in the study will undergo regular evaluation of clinical and para-clinical parameters as described in the study timetable (Section 2.1). This monitoring schedule will vary depending on the choice of the regimen chosen (e.g. more close monitoring of Haemoglobin level for patients taking Linezolid).

The bacteriological monitoring will be the same for all patients regardless of their treatment regimen. Every month throughout treatment, an early morning sputum specimen will be collected. To ensure the availability of culture results at key moments of the treatment, a second specimen will be collected on the spot at the end of month 4 and the end of the treatment. Sputum smear and culture will be done monthly. DST and/or molecular testing will be done at baseline for first- and second-line drugs. In case of sputum non-conversion or reversion, DST will be performed for the same drugs plus Bedaquiline.

The initial culture isolate will be stored at -20°C (or if possible -80°C) in one of the National TB Reference Laboratories for at least 21 months from patient enrolment into the study. In case of recurrence during 12 months after completing treatment, genotyping on initial and recurrent isolates will be performed to allow the distinction between relapse and re-infection.

At each visit, patients will be asked about adverse events (AE) and all answers will be recorded on the Patient's file (*ad hoc* form if patient file is not used routinely). Patients should be encouraged to seek care from study staff if adverse events occur between scheduled visits. If patients present with adverse events or encounter other problems requiring specific investigations between the scheduled intervals, the frequency of monitoring and supervision will be adapted, and the necessary investigations will be repeated as often as required. SAEs and AEs of interest and decisions made during monitoring and supervision will be reported and added to the individual patient records.

11.4 Discontinuation of the treatment regimen

The treatment regimen will be discontinued in some patients. In such cases, patients will be evaluated by their treating centre and switched to an individualized regimen, based on the WHO guidelines for regimen design and after consultation with the BTS MDR-TB CAS. The most common situations in which the regimen may be discontinued include:

- **Resistance to drugs in the shorter MDR/RR-TB regimen.** For patients who submit a sputum sample for culture-based second-line DST at the beginning of treatment, results may not be available until after treatment has started. If resistance to drugs in the original regimen is discovered after treatment is initiated, it may be necessary to modify, extend or discontinue the regimen.
- **Intolerable severe toxicity.** One or more drugs may need to be suspended permanently due to severe toxicity. In such cases, the treating team should review the medical history carefully to determine how the regimen should be modified.
- **Treatment failure.** If clinical and bacteriological responses to treatment are poor (informed by culture result of month 4 or because of a culture reversion), a change in the treatment regimen might be considered. DST should be repeated, regardless of whether the regimen is changed, in order to inform future management decisions.

Clinical Advice Service:

The BTS MDR-TB Clinical Advice Service (BTS MDR-TB CAS) was put in place in 2018 with the intention of facilitating the provision of expert advice on the treatment and monitoring of MDR-TB, increasing the understanding of drug toxicity patterns across the UK, and providing a formal gatekeeping function for the use of specially commissioned and novel drugs, including bedaquiline and delamanid.¹²

11.5 Post-treatment follow-up

After completion of treatment, patients will be informed of the risk of recurrent TB and advised to return for clinical assessment and sputum collection at least at 6 and 12 months after completion of treatment or at any time after the end of the treatment if experiencing TB clinical symptoms.

During the follow-up visits, the following procedures will be undertaken on all patients (regardless of symptoms):

1. Clinical evaluation including evaluation for any adverse events that may have occurred after patient's last visit and any concomitant medications s/he may have received;
2. Sputum collection for smear and culture. A single sputum specimen for smear and culture will be collected at each follow-up visit. DST examination (molecular and/or culture based) and LPA will be done if culture positive. Culture isolate will be stored at -20°C for genotyping.
3. Chest X-ray in case of suspicion of TB recurrence.

In case patients miss their appointment, they will be contacted by phone to check TB clinical symptoms. In case of symptoms consistent with TB, sputum will be collected and sent to the closest TB diagnostic centre for smear and culture and further investigations if a recurrence is suspected.

12. Safety monitoring, management, reporting and recording

Safety monitoring, management and reporting will be the same as the aDSM recommended by WHO.¹⁶

12.1 Safety monitoring

As part of their routine treatment clinic visits, patients will be screened regularly by a person trained in the diagnosis and management of adverse events (AE). An AE is an unexpected medical problem that was not present at baseline and happens (or increased in terms of intensity) during treatment period and up to 6 months after the end of the treatment (this is to take into account the half-life of bedaquiline). AEs may be mild, moderate, or severe, and may be caused by something other than the drug or therapy being given.

12.2 Safety management

One or more drugs may need to be suspended or the dose reduced. Replacement of offending drugs will take the clinical condition and bacteriological status of the patient into account and a decision made after careful case review with the BTS MDR-TB CAS. All AEs will be managed until resolution, which may be after treatment completion.

Management of AEs commonly attributed to the use of conventional MDR/RR-TB drugs is detailed in the Companion Handbook to the WHO Guidelines for the Programmatic Management of Drug-resistant Tuberculosis and the aDSM framework.

12.3 Safety reporting

All serious adverse events will be managed and reported by treating centres according to their usual practice. Treating centres will also be requested to report serious adverse events to the study management group to facilitate real-time pharmacovigilance and updates to treatment centres as the study progresses.

12.4 Safety recording

All SAEs and AE of interest (please see list in Annex 8) will be graded according to the Division of AIDS Table for Grading the Severity of Adult and Paediatric Adverse Events (“DAIDS AE Grading Table”) (please see the common terminology used in Annex 9).

The principal investigator, in collaboration with the study management group will grade and analyse SAEs and AEs for causal relation between the adverse event and one or several drugs and consider possible other causes of the observed AE before concluding that they are due to a particular anti-TB drug. Determination of the category of relationship (causality categories) includes “definite”, “probable”, “possible”, “unlikely”, “not related” and “unclassifiable”. It is shown in Annex 9. Definitions for anticipated or unanticipated ADR are also provided in Annex 9. All information related to SAEs and AEs of interest will be recorded in the study database.

13. Data collection and management

Socio-demographic, clinical, and laboratory data will be collected on all patients onto both routinely used forms and registers, and study specific forms. Clinical data including adverse events will be recorded onto the patient’s medical record and aDSM at each visit. Laboratory data will be extracted from patient’s medical records as well. Data will be abstracted onto the individual patient study case report forms (CRF) at least monthly.

An electronic data entry form (REDCap) that can be used on tablet devices, desktop computers and laptops will be used for data collection. Participating sites will be able to collate and review their own locally collected data for the purposes of audit and quality improvement.

Data collected through the e-form will be directly exported for analysis into a statistical software (e.g. Stata or R), and monitored for completeness and accuracy in real time.

All staff will be trained on the use of the e-forms and data capture. All data will be de-identified to ensure confidentiality. De-identified data will be stored in secure servers at the London School of Hygiene and Tropical Medicine for a minimum of 10 years.

The Chief Investigator will ensure there are adequate quality and number of monitoring activities conducted by the study team. This will include adherence to the protocol, procedures for consenting and ensure adequate data quality. The Chief Investigator will inform the sponsor should he have concerns which have arisen from monitoring activities or if there are problems with oversight or monitoring procedures.

Please see Section 15.2 for further details on protection of confidentiality.

14. Data analysis

14.1 Baseline characteristics of the study population

After the data have undergone cleaning, basic descriptive statistics and cross-tabulations will be produced to display the patient population, model of TB care (place of treatment, number of visits etc.), demographics (e.g. age, gender), socio-economic position, and TB treatment information (e.g. diagnostic delay).

The distribution of categorical variables will be summarised by percentages. Quantitative variables will be summarised using the mean and standard deviation (SD), and/or the median and inter-quartile range (IQR), and the minimum and maximum values. The percentage of patients with missing data will be given, for each demographic and clinical variable.

14.2 Estimating the outcomes of interest

The effectiveness and safety outcomes of interest will be described at each specified point in time (table 1). The outcomes between patients receiving different treatment regimens will be compared using a *chi square* or Fisher's exact test with 95% confidence intervals and a two-sided p value, as appropriate.

14.3 Controlling for confounders

A multivariate analysis with Cox regression will be conducted to control for potential confounders. This will include on-treatment characteristics (culture results, AEs) that could lead to regimen change or termination.

15. Ethical considerations

15.1 Ethics approval

The master protocol of this study has been approved by the Research Ethics Review Committee of the WHO. This adaptation for the UK setting will be submitted for ethical review by the NHS Health Research Authority (HRA) and the London School of Hygiene and Tropical Medicine (LSHTM) Research Ethics Committee (REC). Following centralized approval by the HRA, processes required for implementation at each participating site will be followed.

15.2 Protection of confidentiality and patient protection

Protection of patient confidentiality is essential, and the study will follow the principles of the 2018 Declaration of Helsinki. No patient will be enrolled into this study until he/she (or the guardian in the case of children) has provided written informed consent.

The patient file that will be used to input data into the CRF will be kept in the study site in a secure place. The data collection instrument (CRF) does not include a name or other data from which an individual can be identified. Instead, a patient number will be generated. This will be used to allow linkage back to the register and medical records in case that is needed for quality control, validation of data or a collection of treatment outcome data for study participants. Only authorized study staff will have access to the patient medical record. All study staff will be trained in principles of Good Clinical Practice before the study commences.

15.3 Informed consent

Patients who are eligible for inclusion in the study will be given information about drug-resistant TB and the range of possible treatment regimens. Patients will be provided with information in English, Welsh, Urdu, Hindi, Romanian or Russian language, if requested. Consent for enrolment will be based on a Patient Information Sheet.

Patients will have the opportunity to discuss the Patient Information Sheet with the research staff and will be provided with sufficient time to decide whether they wish to participate or not. Patients will be assured that their decision to participate in the study or not will not affect the quality of care they will receive. Once the patient agrees to participate in the study, the patient will be asked to sign the consent form (or give a thumb print in the presence of a witness, if illiterate). For children, consent from the guardian will be sought as well as assent of the children. The person consenting the patient will be a dedicated person (e.g. data collector) who will be trained to be able to explain the study (as written in the patient information sheet) and document the patient's consent (in the informed consent form).

All patients who are not eligible for the study, or refuse to be enrolled, or withdraw after enrolment, will be managed according to national guidelines with no negative consequences for the patient.

16. Administrative considerations

16.1 Study governance

This research will be conducted in compliance with LSHTM's Good Research Practice Policy and national guidelines, including the Research Councils UK and the UK Research Integrity Office.

This study protocol, which has been adapted for the UK context by clinicians from the community of participating sites (NHS Trusts), will be delivered at multiple TB clinics. All sites will be invited to propose 1-3 named individuals who will serve as site representatives in the TB ShORRT UK Consortium. A Consortium agreement has been drawn up and agreed between participating sites, establishing expectations and aligning understandings of participants with regards to data sharing, study and sub-study leadership, budget allocation (if funding support is secured), publication plans and any other relevant study-related issues. A core study management group of 6-12 representatives drawn from the Consortium members will be responsible for study oversight. Initially this core group will be led by the co-principal investigators, however the leadership of the core group will subsequently rotate.

The co-principal investigators will ensure that all the staff participating in this study have been adequately trained and provided with the resources needed to undertake this research.

For the purposes of transparency and good record-keeping, all formal discussion/decision making meetings (including those concerning budget and publication) will be minuted.

16.2 Public and Patient Involvement (PPI)

Former and current patients with drug-resistant TB have assisted with the development of the study protocol and accompanying documents.

If funding permits, a formal PPI advisory group will be convened for the duration of the study, with a plan to meet at least every 6 months to review recruitment numbers, troubleshoot issues arising at study sites and review interim data. The PPI advisory group will convene under the supervision of an experienced PPI lead who will provide training to PPI members. Two rotating members from the PPI advisory group will also sit on the Study Management Group. PPI members will assist with dissemination of study findings on completion of the study.

16.3 Study team

Investigator's name	Role within the study	Group/ Institution name	Country
David Moore	Chief Investigator	London School of Hygiene and Tropical Medicine	UK
Peter Crook	Co-investigator	St George's, University of London	UK
Sedona Sweeney	Co-investigator	London School of Hygiene and Tropical Medicine	UK
Annabelle South	Co-investigator	University College London	UK
Conor Tweed	Co-investigator	University College London	UK
Duncan Barron	Co-investigator	Freelance	UK
Heinke Kunst	Collaborator	Barts Health NHS trust	UK
Simon Tiberi	Collaborator	Barts Health NHS trust	UK
Effrossyni Gkrania-Klotsas	Collaborator	Cambridge University Hospitals NHS Foundation Trust	UK
Georgina Russell	Collaborator	Cambridge University Hospitals NHS Foundation Trust	UK
Jonathan Underwood	Collaborator	Cardiff and Vale UHB	UK

Faustina Annan-Addae	Collaborator	Chelsea & Westminster Hospital NHS Foundation Trust	UK
Helen Burgess	Collaborator	Chelsea & Westminster Hospital NHS Foundation Trust	UK
Patrick Lillie	Collaborator	Hull University Teaching Hospitals NHS Trust	UK
Onn Min Kon	Collaborator	Imperial College Healthcare NHS Trust	UK
Charlotte Hall	Collaborator	Leeds Teaching Hospitals NHS Trust	UK
Padmasayee Papineni	Collaborator	London North West University Healthcare NHS Trust	UK
David Connell	Collaborator	Ninewells Hospital, NHS Tayside	UK
Matthijs Backx	Collaborator	Public Health Wales	UK
Marc Lipman	Collaborator	Royal Free London NHS Foundation Trust	UK
Paul Collini	Collaborator	Sheffield Teaching Hospitals NHS Foundation Trust	UK
Danielle Cohen	Collaborator	Sheffield Teaching Hospitals NHS Foundation Trust	UK
Amber Arnold	Collaborator	St George's University Hospitals NHS Foundation Trust	UK
Tom Harrison	Collaborator	St George's, University of London	UK
Jim Macfarlane	Collaborator	The Newcastle upon Tyne Hospitals NHS Foundation Trust	UK
Santino Capocci	Collaborator	University College London Hospitals NHS Foundation Trust	UK
Stephen Morris-Jones	Collaborator	University College London Hospitals NHS Foundation Trust	UK
Philip Gothard	Collaborator	University College London Hospitals NHS Foundation Trust	UK
Hanif Esmail	Collaborator	University College London Hospitals NHS Foundation Trust	UK
Michael Brown	Collaborator	University College London Hospitals NHS Foundation Trust	UK
Mahdad Noursadeghi	Collaborator	University College London Hospitals NHS Foundation Trust	UK
Martin Dedicoat	Collaborator	University Hospitals Birmingham NHS Foundation Trust	UK
Sarah Mungall	Collaborator	University Hospitals Bristol and Western NHS Foundation Trust	UK
Gerry Davies	Collaborator	University of Liverpool; Liverpool School of Tropical Medicine	UK
Derek Sloan	Collaborator	University of St Andrew's	UK
Lusha Kellgren	Collaborator	Whittington Health NHS Trust	UK
Moerida Belton	Collaborator	Whittington Health NHS Trust	UK

17. Data ownership and sharing

All patient records will be deidentified to ensure confidentiality. All records will be encrypted and stored at the London School of Hygiene and Tropical Medicine servers for a minimum of ten years, in accordance with LSHTM's Records Retention and Disposal Schedule. The records will be kept in a clear and complete manner, including negative results. These records will be publicly available and will allow for retrospective or prospective audits and access for research purposes.

The results of the study will be disseminated via scientific conferences, international peer-reviewed journal publications and technical reports. An online annual report of important findings will be published and made available for the public to consult.

De-identified data may be shared with WHO or other international partners seeking to collate TB-ShORRT datasets into a multi-national individual patient data analysis, according to provisions made in the Consortium agreement or any subsequent Consortium decision.

18. Study budget

Funding will be sought for the study as a whole and / or for individual work packages if required.

PROTOCOL - PART B

Work Packages 2, 3 and 4: Evaluating the acceptability, costs and the impact on health-related quality of life of second-line treatments for tuberculosis in the United Kingdom

Part B of this protocol describes Work Packages (WP) 2, 3 and 4

- WP2a Impact on health-related quality of life (HRQoL): **Section 2**
- WP2b Stigma: **Section 3**
- WP3: Acceptability: **Section 4**
- WP4 Cost-effectiveness: **Section 5**

1. Foreword

While the programmatic use of newer second-line treatment regimens for TB under operational research conditions can provide important data to the global TB community about their effectiveness and safety, it can also provide information about other important outcomes, such as health-related quality of life (HRQoL) and socioeconomic impact for patients and households, as well as programmatic implementation.

Quantitative, qualitative and mixed-method research approaches can be used to gain a deeper understanding of the implications that second-line regimens can have on the quality of life of patients with TB (WP 2a), including stigma and discrimination they may experience as a result of their condition (WP 2b).

In order to have an impact, the intervention must not only be effective and safe, but must be feasible and acceptable to the local population and to the health care staff involved in its delivery. WP 3 describes qualitative operational research into acceptability of the implementation of novel all-oral shorter regimens to health and social service care workers.

Complementary operational research may also include cost-effectiveness analysis, both from patient's and provider's perspective (WP 4).

The adoption of longitudinal add-on studies alongside the implementation of novel TB treatment regimens provides the opportunity to repeat measures and make comparisons over time, and even describe changes that extend beyond treatment completion.

Part B – Work Packages 2, 3 and 4

2. WP2a - Health-related quality of life (HRQoL)

2.1 Background

Quality of life refers to a subjective evaluation of the effects of disease and health interventions, which is embedded in a cultural, social and environmental context. HRQoL measures the physical and functional status, and social and emotional well-being of an individual.

Although limited, previous studies have shown that HRQoL varies among TB patients at different stages of treatment and is significantly worse than HRQoL among the general population at all stages of treatment. After completing TB treatment, HRQoL among TB patients improves significantly compared with pre-treatment baseline; however, HRQoL remains lower than in the general population in many domains, even beyond one year of completing treatment. Evidence has also shown that HRQoL among MDR/RR-TB patients, who by definition require second-line medications, is worse compared to drug-susceptible TB.

It is reasonable to assume that individuals treated for drug-resistant TB with novel, all-oral shorter regimens would experience better quality of life compared to patients on a standard (either short or long) MDR/RR-TB regimen, through two primary mechanisms. First, an injectable-free treatment itself can potentially improve treatment adherence of patients and facilitate the implementation of community-based models of care. This in turn would reduce direct costs incurred by the patients associated with travelling to the health facility for daily injections, and/or the opportunity costs associated with lost productivity and income due to hospitalisation in the intensive phase of treatment.

Second, injectable-free treatment regimens that potentially reduce the occurrence of adverse reactions would likely improve adherence to treatment, and lead to less HRQoL decrement. Patients treated with shorter regimens may also return to usual activities faster. It is therefore likely that overall health-related quality of life of patients would improve.

Implementation research into HRQoL therefore provides important complementary information related to the implementation and uptake of novel second-line treatment regimens.

2.2 Study objective

The objective of this study is to assess the HRQoL of patients receiving non-first-line TB regimens, in particular those with MDR/RR-TB patients receiving the all-oral shorter MDR/RR-TB treatment regimens.

It is anticipated that study centres may gradually adopt novel regimens (for example those containing pretomanid if this becomes available); if this occurs, comparisons can be drawn between shorter course regimens compared to current regimens of ≥ 9 months.

2.3 Study design

This study investigates HRQoL among TB patients receiving non-first-line treatment regimens, including short and long course regimens and regimens both with and without injectable agents (Figure 3).

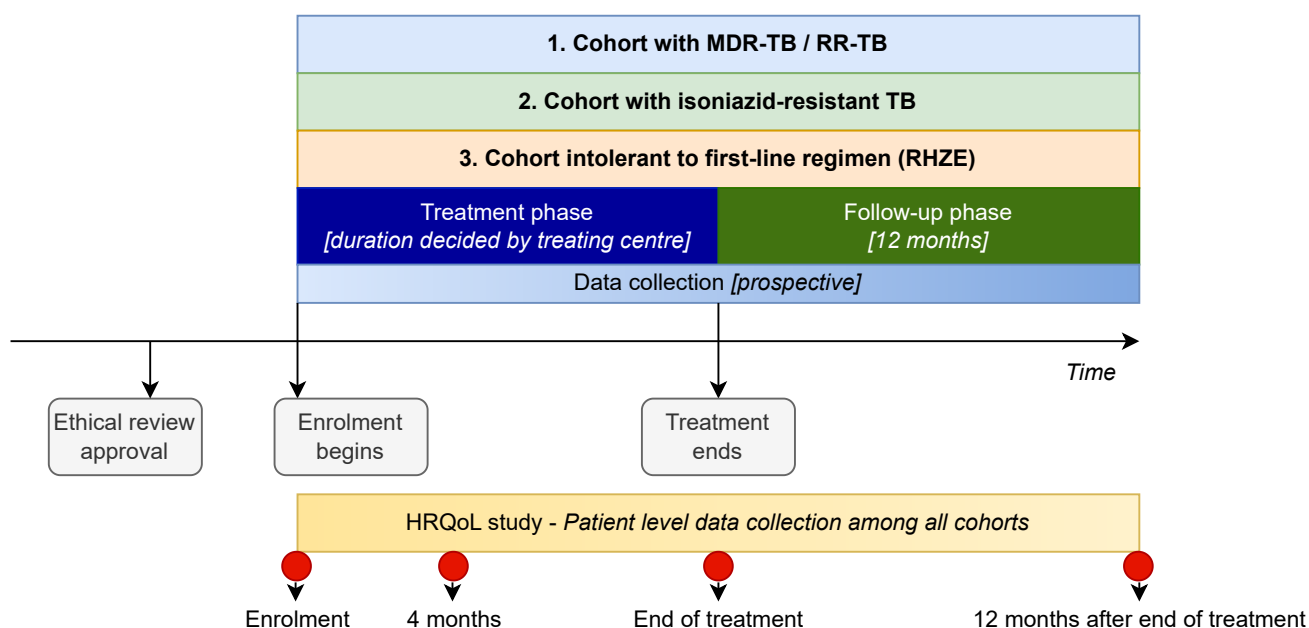


Figure 3: Design for the conduct of HRQoL study

This assessment adopts a longitudinal design with data collection at enrolment, at 4 months, at end of treatment, and twelve months after end of treatment, time points that are intended to capture health-related quality of life at diagnosis, during treatment, at the end of treatment, and long-term follow-up once treatment has been completed.

This study adopts a semi-qualitative approach, employing scales to assign scores to responses from patients, and interview techniques to explore the motivations for these scores in more detail.

2.4 Study population

The study will be conducted on patients receiving non-first-line treatment regimens for TB due to either resistance patterns or drug intolerance. All eligible patients (as per Section 7 of part A of this protocol) will be invited to complete the questionnaire for this sub-study at the time points stated above, and to participate in a structured interview after completing treatment.

2.5 Study procedures

Patients will complete the EQ-5D-5L and FACIT-TB questionnaires and the results will be transcribed into the REDCap database by the study team.

2.6 Measurement instruments

The evaluation of health-related quality of life will utilise the EQ-5D-5L and FACIT-TB measurement tools. The EQ-5D-5L comprises five dimensions: mobility, self-care, usual activities, pain/discomfort and anxiety/depression, with 5 levels assigned to each dimension: no problems, slight problems, moderate problems, severe problems and extreme problems.¹⁷

We will also be using the FACIT-TB questionnaire,¹⁸ which was the best-performing tool in a multi-country study evaluating patient-centred outcome measures for DR-TB clinical trials, in terms of appropriateness, reliability, responsiveness, precision, acceptability and feasibility. FACIT-TB assesses physical, social/family, emotional, functional and spiritual well-being using a Likert-type scale for items ranging from “not at all” (score 0) to “very much” (score 5).

Participants will be invited to take part in structured interviews after completing treatment to explore the dimensions measured by the questionnaires in more detail.

2.7 Data analysis

The distribution of categorical variables will be summarised by percentages. Quantitative variables will be summarised using the mean and standard deviation (SD), and/or the median and inter-quartile range (IQR), and the minimum and maximum values. The percentage of patients with missing data will be given, for each demographic and clinical variable.

Scores for EQ-5D-5L and FACIT-TB at baseline, 4 months, end of treatment, and 12 months post treatment, and percentage change in score over the treatment period will be compared using means, mean differences and t-test, with a two-sided p-value. Scores generated for each dimension being assessed by the measurement tools will be compared as well as the total score for FACIT-TB and overall state of health score for EQ-5D-5L.

Qualitative analysis of structured interviews will involve the generation of themes and the use of a thematic network method. Findings will be reported using Consolidated Criteria for Reporting Qualitative Research.

3. WP2b - Measuring stigma

3.1 Background

TB-related stigma is an important social determinant of health which can have a considerable impact on individuals and communities, including delays in seeking health care and the ability to manage illness and complete treatment.

In the UK, TB is concentrated in groups often characterized by complex health and social risks that are associated with stigma, for example, homelessness, imprisonment, substance misuse, HIV, and recent history of migration. These groups suffer from TB-related stigma either due to illness itself or particular practices of TB control measures and media reporting of TB.¹⁹

Persons with DR-TB may be uniquely exposed to and disadvantaged by stigma, and their experiences of stigma often differ markedly from patients with drug susceptible TB. This is due to the following reasons:

1. DR-TB treatment takes longer than DS-TB treatment, and therefore the exposure to stigma may be longer as the identity of “TB patient” is less transient.
2. Second-line treatments are typically more toxic, with more side effects. People on second-line treatments are more likely to experience neuropsychiatric or perception altering side effects as a consequence of their comprehensive treatment regimens and to face catastrophic costs due to the length of their treatment, both of which can heighten their vulnerability to stigma. Hearing loss, psychological side effects, and impoverishment can reinforce the social construction of DR-TB patients being deviant, unpredictable, and dangerous.
3. DR-TB is often assumed to be caused by misbehaviour. Unlike in cases of drug-susceptible TB, in the case of DR-TB, there may be treatment adherence behaviours that may contribute to the development of acquired DR-TB. This may tempt health workers to blame individuals for their disease. DR-TB clients may be at higher risk of self-stigma if they harbour self-blame or guilt related to drug resistance caused by non-adherence.
4. DR-TB has more potential to create fear. One of the main facets of all stigma constructs is the social construction of people with TB as being dangerous to the wider community. Perceptions of TB curability is associated with lower levels of TB stigma. When a person has DR-TB, doubts about curability may fuel the notion that DR-TB patients represent a mortal risk to others. Therefore, DR-TB may have a greater ‘mark’ than drug-susceptible TB.

TB stigma extends beyond the denigration of people with TB/DR-TB. It can likewise harm those who may not have TB themselves (such as family members, friends, volunteer caregivers, HCWs, TB activists, and minors), but are associated with the disease through their work or relationships (so-called “secondary TB stigma”). For example, TB health care workers can experience stigmatization and discrimination due to being in close contact with TB patients (so-called “dirty work stigma”). Second-line treatments themselves may cause stigma because of visible signs such as skin pigment change due to clofazimine.

3.2 Study objective

The objective of this study is to assess TB-related stigma among TB patients receiving second-line treatment regimens.

3.3 Study design

This study will employ a mixed-method approach. Prevalence of stigma will be measured through a quantitative approach (patient survey) employing scales.

In the context of this study, stigma will be investigated among TB patients receiving non-first-line treatment regimens. (Figure 4).

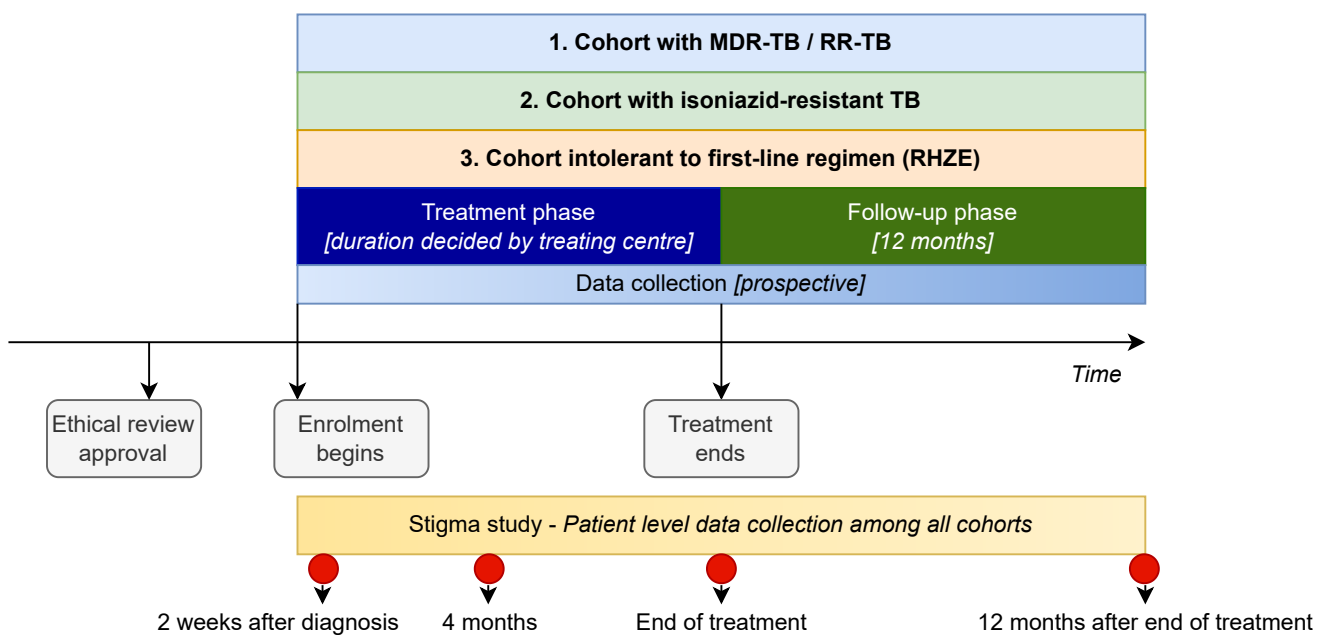


Figure 4: Design for the conduct of the stigma study

3.4 Study population

The study population will include all patients with drug-resistant tuberculosis, as described on section 7 of Part A of the protocol.

This assessment adopts a longitudinal design with data collection at two weeks after TB diagnosis, at 4 months, at end of treatment, and twelve months after end of treatment, time points that are intended to capture stigma soon after diagnosis, during treatment, at the end of treatment, and long-term follow-up once treatment has been completed.

3.5 Measurement tools

We will investigate the impact of treatment on disease-related stigma using the Stigma Scale for Chronic Illness, which has been previously studied in TB, and the Cataldo Lung Cancer Scale, which has been adapted to measure stigma among people with MDR/RR-TB.²⁰ There will be structured interviews with study participants after completing treatment to explore the relationship between disease-related stigma and treatment in more detail.

3.6 Data analysis

Stigma scores at 2 weeks, 4 months, end of treatment, and 12 months post-treatment, and percentage change in score over the treatment period will be compared using means, mean differences and t-test, with a two-sided p-value.

Qualitative analysis of structured interviews will involve the generation of themes and the use of thematic network method. Findings will be reported using Consolidated Criteria for Reporting Qualitative Research.

Part B – Section 4

4. WP3 - Acceptability of model of care to health care providers

4.1 Background

It is important to ascertain attitudes and perceptions of health care providers who are involved in patient care as this can inform how to improve patient support services and the model of care related to the adoption of novel second-line regimens.

4.2 Work package 3 objective

The objective of this work package is therefore to capture and understand the perspectives of healthcare providers with a focus on identifying potential barriers to adoption associated with the delivery of this intervention. This work package will address two research questions:

1. What do health care providers perceive as barriers to uptake of short course regimens for people with drug-resistant tuberculosis in the UK?
2. What strategies or facilitators might encourage health care providers to use short course regimens for people with drug-resistant tuberculosis?

4.3 Theoretical frameworks

There is a substantial theoretical literature on knowledge translation. Our qualitative study will be informed by the following theoretical frameworks:

- Theoretical Domains Framework:²¹ this framework provides a comprehensive, theory-informed approach to identifying the determinants of behaviour. The domains included in the framework are: knowledge; skills; social/professional role and identity; beliefs about capabilities; optimism; beliefs about consequences; reinforcement; intentions; goals; memory, attention and decision processes; environmental context and resources; social influences; emotion; behavioural regulation.
- Framework for Knowledge Translation:²² this framework was developed to help those involved in knowledge translation better understand user groups. The framework consists for five domains: the user group; the issue; the research; the knowledge translation relationship; and dissemination strategies.
- Research Development, Dissemination and Utilisation Framework:²³ this framework identifies seven factors that help explain dissemination and utilisation: linkage; structure of resource system, user system & dissemination-utilisation strategy; openness; capacity; reward; proximity and synergy.

Using these frameworks will ensure that our analysis builds on what is already known about the knowledge translation process and what influences it. These frameworks will initially be used to design the topic guide for semi-structured interviews. One or more of them will then be used in the design of the framework for analysis, depending on which fits best with the data we gather.

4.4 Study population

Our study population is healthcare workers involved in treating people with DR-TB at sites participating in the TB ShORRT study in the UK. We will include doctors, nurses and DOT workers. We will use a purposive sampling strategy, to try to get a sample that encompasses diversity in terms of:

- *Role (doctor – consultant / doctor – specialty registrar / TB nurse / DOT worker)
- *Number of people with DR-TB they have treated in the last year (<4, ≥4)
- Whether they are a member of the MDR-TB CAS (yes or no)
- Type of centre (academic / non-academic)

The starred criteria will be our primary sampling criteria, while the other two will be secondary. Tertiary criteria that will be monitored as recruitment continues include gender and, for doctors, whether their specialty is infection or respiratory.

Sampling matrix

		Treated <4 DR-TB cases in last year	Treated ≥4 DR-TB cases in last year
Role	Consultant	Minimum (Min). 2	Min. 2
	Specialty registrar	Min. 1	Min. 1
	TB nurse	Min. 2	Min. 2
	DOT worker	Min. 1	Min. 1
Member of MDR-TB CAS?	Yes	No min	Min. 2
	No	Min. 2	No min
Centre	Academic	Min. 2	
	Non-academic	Min. 2	
Gender	Male	No min.	
	Female	No min.	
Specialty (for doctors)	Respiratory	No min.	
	Infection	No min.	

4.4.1 Sample size

Using the Information Power model²⁴ to identify the likely sample size required to answer our research questions, we anticipate having good information power, requiring a moderate sample of 12-16 participants. We will repeat our appraisal of information power along the process, as we find out more about the quality of dialogue and sample specificity.

Information Power considerations:

- **Study aim:** reasonably narrow
- **Sample specificity:** dense, as participants will have specific experience that is relevant to the study aim (experience of treating patients with DR-TB)
- **Established theory:** this study will be using established theoretical frameworks, as discussed above.
- **Quality of dialogue:** the interviewer for this study has experience of interviewing UK health professionals, and has done previous research in the area of DR-TB, so we expect the quality of dialogue to be moderate or good, depending on whether participants are able to spare sufficient time for the interviews.
- **Analysis strategy:** we plan to conduct both cross- and within-case analysis.

4.5 Study design

This study is a longitudinal qualitative study, with data collected via semi-structured interviews with health care providers. We plan to conduct two rounds of qualitative data collection, with the first round taking place during the first year of the study. People who participate in the first round of interviews will be invited to take part in the second round of interviews, 12-18 months after their initial interview (although this is optional). This will allow us to explore how individuals' views on short course treatments for DR-TB evolve over the course of the study, as their experience of these regimens, and the context in which they are working, changes.

4.6 Recruitment of participants

We will ask PIs at participating sites to share details of the qualitative study with health care providers providing care to people receiving non-first line TB therapy. Those interested in taking part in the qualitative research will be given a link to our Participant Information Sheet and electronic consent form (in REDCap), and, if they consent, invited to complete a short survey (also in REDCap) to gather information relating to our purposive sampling criteria (see section 4.4), and contact details. This electronic consent form will be created in REDCap within the UCL

Data Safe Haven to ensure that potentially identifiable information is stored directly in a secure environment. We will then contact those who fill gaps in our sampling frame to arrange interviews.

Participants will be asked to consent to audio-visual recordings being made and for quotations (anonymised) to be used in future publications. Participants will be informed that anonymised data may be shared with other researchers in future.

4.7 Data collection

Data about participants' characteristics (as outlined in section 4.4) will be collected using a short survey in REDCap, stored on the UCL Data Safe Haven. Qualitative data will then be collected via semi-structured interviews carried out via Microsoft Teams. People who participate in the first round of interviews will be invited to take part in the second round of interviews 12-18 months later. Interviews will be audio-visually recorded in Teams, with a back-up audio recording being made on an encrypted iPad. The recording will be transcribed with the help of Microsoft Teams' automatic transcription facility.

The semi-structured interviews will be guided by a topic guide, with questions about the interviewee's views on short course treatments for drug-resistant TB, and the barriers and facilitators to their use in the UK. The questions will be informed by the domains and factors identified in the theoretical frameworks summarised in Section 4.3. The topic guide will be revised as the interviews progress, learning from the experience of previous interviews, allowing us to explore issues that come up in previous interviews in more depth. During the interviews, participants will be able to skip questions, take breaks or end the interview at any time. We anticipate the interviews generally lasting between 30-60 minutes.

4.8 Data management and storage

Informed Consent forms and details from the short questionnaire about participant characteristics will be collected via REDCap and stored in the UCL Data Safe Haven. The interview recordings and transcripts will also be stored in the UCL Data Safe Haven. The Data Safe Haven has been certified to the ISO27001 information security standard and conforms to NHS Digital's Information Governance Toolkit. Built using a walled garden approach, where the data is stored, processed and managed within the security of the system.

Automatically-generated transcriptions will be checked against the original recordings and corrected as required. The transcripts will then be pseudonymised (including removing information such as names of individuals or hospitals) to prevent identification of the interviewee. Only the lead researcher in this work package will have access to the recordings and non-pseudonymised transcripts. Pseudonymised transcripts will be stored for analysis on UCL servers, with access only granted to members of the team involved in this aspect of the research.

4.9 Analysis

As our research is building upon established theoretical frameworks around knowledge translation, we will use a Framework Analysis approach to analyse our data.²⁵ The stages of the analysis will be:

1. Familiarisation with the data
2. Construction of the initial thematic framework
3. Indexing and sorting the data using this framework
4. Review data extracts for coherence & further refinement of the framework
5. Data summary and display in the form of a matrix, with each interview forming a row, and each column representing a different theme or sub-theme from the framework

We will use computer-assisted analysis software (NVivo 12 and/or Atlas.ti 23) to manage the indexing, sorting and display of the data.

Analysis will be carried out alongside data collection (i.e. we will not wait until we have collected all the qualitative data to start analysing). Carrying out analysis while data collection continues will allow us to adapt the topic guide as needed to explore issues of interest that arise in previous interviews or through our analysis.

4.10 Ethical considerations

Given the subject matter and study population of this study, we do not anticipate the interviews raising particularly sensitive or upsetting issues for participants. However, as part of the informed consent process we will make sure participants understand the focus on the interviews before deciding whether or not to take part. We will also make sure they understand that participation is voluntary, and that they can choose not to answer any questions they do not want to, and end the interview at any point. Participants who take part in the first round of interviews are not obliged to take part in the second round of interviews.

In order to allow participants to raise issues or views that they may feel are controversial or likely to be unpopular with others, we will ensure that any quotes used from interviewees do not allow identification of the participant, through adequate pseudonymisation. Only the lead of this work package will have access to the recordings and unpseudonymised transcripts.

Part B – Section 5

5. WP4 - Cost and cost-effectiveness analysis

5.1 Background

Nationally representative surveys of costs faced by TB patients and their households are finding consistently higher costs incurred by MDR/RR-TB patients compared to drug-susceptible TB patients, and society at large. As these higher costs are mainly due to the longer duration of treatment, more ADR, and associated number of visits to the health facilities, it is reasonable to expect that a shorter treatment regimen would reduce the number of clinic visits, thus resulting in net cost savings both to the health system and patients.

Health economic analyses can adopt a health system perspective by including only health system costs or a societal perspective by also including patient costs, both as part of the cost inputs and as an outcome in its own right. In addition, cost-effectiveness (with favourable treatment as outcome) and cost-utility analysis (with QALYs as a complementary outcome to favourable treatment) will enable the comparison between new all-oral shorter drug-resistant treatment regimens and older, longer course treatment regimens, as well as with investments non-related to TB.

Here we plan to consider only health system costs, to generate evidence on incremental cost-effectiveness which can contribute to national level (NICE) guidance development.

5.2 Study objective

The objectives of this study are to assess the:

- Total and incremental cost of all-oral shorter drug-resistant TB treatment regimens from the perspective of the NHS and personal and social services.
- Incremental cost-effectiveness (favourable treatment as outcome) of all-oral shorter drug-resistant TB treatment regimens compared to longer course regimens.

Probability of all-oral shorter drug-resistant TB treatment regimens being cost-effective at the current NICE-defined willingness to pay threshold of unit cost per QALY saved

5.3 Study design

The incremental cost-effectiveness of novel regimens as compared with the current standard of care will be evaluated. The costs of each treatment alternative from the provider perspective will be assessed using study event data (WP1) and UK reference pricing, in line with NICE technology appraisal guidance.

The population of interest will be adults receiving treatment for drug-resistant TB at participating health facilities. The sample size for the cost-effectiveness analysis will mirror the sample size for the broader study.

The primary outcome of the analysis will be the incremental cost incurred per QALY gained (using data from WP2) with the intervention regimen compared to the standard of care. The analysis will utilize a Markov model to estimate the probability of new regimens being cost-effective beyond the time horizon of the trial for a setting-specific cohort of MDR TB patients.

5.4 Data collection

Data will be collected from various sources (trial, literature, national databases, questionnaires). Four types of data will need to be collected: 1) outcome data; 2) event and service use data; 3) cost data related to TB. Costs and health impact of health issues unrelated to TB will be excluded from the analysis.

Event data for all study participants will be collected on number, reason, and length of stay for each hospital visit during their treatment episode and follow-up period. We will capture mean number of outpatient visits, inpatient bed-days, monitoring tests, and duration and outcomes for serious adverse events. We will review the published literature on probabilities of relapse and life expectancy after the follow-up period ends.

We will estimate average costs per patient for all services related to TB, received during and after the treatment episode. Where possible we will use healthcare resource groups to estimate resource use, to avoid the need for micro-costing. We will evaluate patient records to identify quantities of drugs, consumables, lab tests for monitoring received throughout the patient episode. The public list prices for technologies will be used to estimate unit costs.

We will include the costs of any additional, short-term technical assistance (such as clinical trainings) allocated to implement the all-oral shorter regimens. These costs of technical assistance will be assessed as capital costs required for establishing the study regimen and as costs for recurrent costs for sustaining it.

A random selection of ten patients will be carried out to review their charts for healthcare utilisation (number of visits, diagnostic and monitoring tests done, etc.) to validate how representative the costs calculated “per protocol” are.

5.5 Data analysis

Total and disaggregated costs to the NHS per patient episode of care will be calculated.

To estimate cost-effectiveness of shorter regimens as compared with longer regimens, we will extrapolate costs and health benefits over a lifetime horizon using a Markov model representing progression of a cohort of patients through different health states and activities. The treatment effects on transmission of MDR-TB will not be modelled. The model will be parameterised with a combination of data from the current study, as well as data from the literature. Expert opinion will be used where no data is available, for example estimating the extent to which patients return to care after loss to follow-up. All model parameters will be reviewed by a panel of local experts.

We will estimate incremental costs, cost per death averted, and cost per quality-adjusted life year (QALY) gained of all-oral shorter drug-resistant TB treatment regimens as compared with conventional longer regimens. Discount rates of 3.5% will be used to discount both costs and effects in the primary analysis, following NICE reference case recommendations. We will vary these rates in sensitivity analyses. We will conduct univariate and probabilistic sensitivity analysis to explore the effects of sampling uncertainty for the estimated incremental cost and incremental effectiveness parameters, together with the impact of methodological assumptions (such as discount rate, study perspective). Where applicable, we will report differences in costs, outcomes, or cost-effectiveness that can be explained by variations between subgroups of patients with different baseline characteristics.

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Annex 1. Treatment regimens tested in selected ongoing clinical trials

Note: The use of drug regimens where Pretomanid is combined with other drugs than in the BPAL regimen warrants more rigorous research conditions than operational/implementation research implies.

Clinical trial	Regimen	Ongoing / completed	All drugs are commercially available	FQ-S/FQ-R
STREAM 1 regimen B	Cfz, E, Z, Mfx, H, Km (16 weeks); followed by Cfz, E, Z, Mfx (24 weeks)	Enrolment completed	Yes	FQ-S
NiX-TB	Bdq, Pa, Lzd (24-36 weeks)	Enrolment completed	No	XDR
MDR END	Dlm, Lzd, Lfx, Z (36-52 weeks)	Enrolling	Yes	FQ-S
STREAM 2 regimen C	Bdq, Cfz, E, Z, Lfx, H, Pto (16 weeks); followed by Bdq, Cfz, E, Z, Lfx (24 weeks)	Enrolling	Yes	FQ-S
STREAM 2 regimen D	Bdq, Cfz, Z, Lfx, H, Km (8 weeks); followed by Bdq, Cfz, Z, Lfx (20 weeks)	Enrolling	Yes	FQ-S
PRACTECAL regimen 1	Bdq, Pa, Lzd (24 weeks)	Enrolling	No	FQ-S and R
PRACTECAL regimen 2	Bdq, Pa, Lzd, Cfz (24- weeks)	Enrolling	No	FQ-S and R
PRACTECAL regimen 3	Bdq, Pa, Lzd, Mfx (24-weeks)	Enrolling	No	FQ-S and R
endTB regimen 1	Bdq, Lzd, Mfx, Z (39 weeks)	Enrolling	Yes	FQ-S and R
endTB regimen 2	Bdq, Cfz, Lzd, Lfx, Z (39 weeks)	Enrolling	Yes	FQ-S and R
endTB regimen 3	Bdq, Dlm, Lzd, Lfx, Z (39 weeks)	Enrolling	Yes	FQ-S and R
endTB regimen 4	Dlm, Cfz, Lzd, Lfx, Z (39 weeks)	Enrolling	Yes	FQ-S and R
endTB regimen 5	Dlm, Cfz, Mfx, Z (39 weeks)	Enrolling	Yes	FQ-S and R

Observational research				
STREAM adapted regimen with Linezolid	4-6 months of Linezolid + Moxifloxacin + Prothionamide + Clofazimine + Pyrazinamide + High-dose Isoniazide + Ethambutol, followed by 5 months of Moxifloxacin + Clofazimine + Pyrazinamide + Ethambutol	Enrolling	Yes	FQ-S

Annex 2. Dosing of medicines used in second-line MDR/RR-TB regimens by weight band in patients older than 14 years

Source: 2019 WHO consolidated guidelines on drug-resistant tuberculosis treatment

Group	Medicine	Weight-based daily dose	Formulation	Weight bands for patients older than 14 years ^a					Usual upper daily dose ^b	Comments
				30–35 kg	36–45 kg	46–55 kg	56–70 kg	>70 kg		
A	Levofloxacin	- ^c	250 mg tab	3	3	4	4	4	1.5 g	
			500 mg tab	1.5	1.5	2	2	2		
			750 mg tab	1	1	1.5	1.5	1.5		
	Moxifloxacin	standard dose ^{c,d}	400 mg tab	1	1	1	1	1	400 mg	As used in the standardized shorter MDR/RR-TB regimen
		high dose ^{c,d}	400 mg tab	1 or 1.5	1.5	1.5 or 2	2	2	800 mg	
	Bedaquiline	- ^c	100 mg tab	4 tabs od for first 2 weeks; then 2 tabs od M/W/F for 22 weeks					400 mg	
	Linezolid	- ^c	600 mg tab	(<15 y)	(<15 y)	1	1	1	1.2 g	
B	Clofazimine	- ^c	50 mg cap or tab ^h	2	2	2	2	2	100 mg	
			100 mg cap or tab ^h	1	1	1	1	1	100 mg	
	Cycloserine or terizidone	10–15 mg/kg	250 mg cap	2	2	3	3	3	1 g	
C	Ethambutol	15–25 mg/kg	400 mg tab	2	2	3	3	3	-	
	Delamanid	- ^c	50 mg tab	2 bd	2 bd	2 bd	2 bd	2 bd	200 mg	

	Pyrazinamide	20–30 mg/kg	400 mg tab	3	4	4	4	5	-	
			500 mg tab	2	3	3	3	4	-	
	Imipenem-cilastatin	– ^c	500 mg + 500 mg powder for injection, vial (10 ml)	2 vials (1 g + 1 g) bd					-	To be used with clavulanic acid
	Meropenem	– ^c	1 g powder for injection, vial (20 ml)	1 vial 3 times per day or 2 vials bd					-	To be used with clavulanic acid
	Amikacin	15–20 mg/kg	500 mg/2 ml solution for injection, ampoule ^g	2.5 ml	3 ml	3 to 4 ml	4 ml	4 ml	1 g	
	Streptomycin	12–18 mg/kg	1 g powder for injection, vial ^g	Calculate according to the dilution used					1 g	
	Ethionamide or prothionamide	15–20 mg/kg	250 mg tablet	2	2	3	3	4	1 g	Once daily dose advised but can start with 2 divided doses until tolerance improves
Other medicines ^f	<i>p</i> -aminosalicylic acid	8–12 g/day in 2–3 divided doses	PAS sodium salt (equivalent to 4 g PAS acid) sachet	1 bd	1 bd	1 bd	1 bd	1 to 1.5 bd	12 g	
			PAS acid (4 g) sachet	1 bd	1 bd	1 bd	1 bd	1 to 1.5 bd		
	Isoniazid	4–6 mg/kg (standard dose) ^d	300 mg tab	2/3	1	1	1	1	-	100 mg isoniazid tablet can facilitate the administration of certain dosages Pyridoxine given with isoniazid in patients at risk (such as those with HIV, malnutrition)
		10–15 mg/kg (high dose) ^d	300 mg tablet	1.5	1.5	2	2	2		
	Clavulanic acid ^g	– ^c	125 mg clavulanic acid as amoxicillin/ clavulanate, 500 mg/ 125 mg tab ^g	1 bd	1 bd	1 bd	1 bd	1 bd	-	Only to be used with carbapenems
	Kanamycin	15–20 mg/kg	500 mg or 1 g powder for injection, vial (2ml); ^e	2 to 2.5 ml	2.5 to 3 ml	3 to 4 ml	4 ml	4 ml	1 g	M/W/F dosing of aminoglycosides at 25 mg/kg/day may limit

			1 g/ 4 ml solution for injection, ampoule; ^e								toxicity and inconvenience when the injectable agents are used in longer MDR/RR-TB regimens
	Capreomycin	15–20 mg/kg	500 mg or 1 g powder for injection, vial (2 ml) ^e	2.5 ml	3ml	3 to 4 ml	4 ml	4 ml	1 g		
	Gatifloxacin	- ^c	400 mg tab	2	2	2	2	2	800 mg		Currently there is no availability of gatifloxacin on the market. Gatifloxacin is not used in persons <18 years old.
	Thioacetazone	- ^c	150 mg tab	1	1	1	1	1	-		Currently there is no availability of thioacetazone on the market. Thioacetazone is not used in persons <18 years old.

(<15 y) = follow the separate dose schedule for patients younger than 15 years of age; bd = two times a day; cap = capsule; g = gram; im = intramuscular; iv = intravenous; kg = kilogram; ml = millilitre; mg = milligram; M/W/F = Monday, Wednesday, Friday; soln = solution; susp = suspension; tab = tablet

Footnotes

- a Dosages were established by the Guideline Development Group for the *WHO treatment guidelines for rifampicin- and multidrug-resistant tuberculosis, 2018 update* and the WHO Global task force on the pharmacokinetics and pharmacodynamics (PK/PD) of TB medicines and other experts. They are based on the most recent reviews and best practices in the treatment of MDR/RR-TB. For certain agents the dosages were informed by pharmacokinetic modelling results based on the principle of allometric scaling (Anderson BJ, Holford NH. Mechanism-based concepts of size and maturity in pharmacokinetics. *Annu Rev Pharmacol Toxicol* 2008;48:303–32). Due to the pharmacokinetic properties of certain medicines the doses proposed may exceed the mg/kg/day ranges shown here in order to achieve blood concentrations similar to target levels in an average adult patient. In patients <30 kg follow the schedule for <15 year olds unless otherwise indicated. If multiple dose options are given for one weight band select the lower or higher option depending on whether the patient is at the lower or higher limit of the body weight range. Dosing more closely to the target mg/kg/day should be aimed for, and is more feasible with oral or parenteral fluids and when solid forms of different dosages are available. Fractioning of tablets into halves or less should be avoided, if possible. Therapeutic drug monitoring is advised when the dose is at the upper and lower ends of the range to minimize the adverse therapeutic consequences of over- and under-exposure, respectively (especially for injectable agents, linezolid and fluoroquinolones).
- b Clinicians may decide to exceed these values in particular cases to improve therapeutic effect.
- c No weight-based dosing is proposed.
- d Unless there is risk of toxicity, the high dose may be used if antimicrobial levels may be lowered because of pharmacokinetic interactions, malabsorption or other metabolic reasons or if the strain has low-level drug resistance.

e Weight-based daily dose is for 6 or 7 days/week administration (M/W/F scheduling may permit higher dosing). Volumes shown may differ by preparation. Streptomycin may be diluted in three different ways. For iv use, the volume may be increased.

f In the 2018 WHO treatment guidelines, these agents are either no longer recommended (kanamycin, capreomycin), only recommended as a companion agent (amoxicillin/clavulanic acid) or not included because of lack of data from the latest analysis on longer MDR/RR-TB regimens in adults (gatifloxacin, isoniazid and thioacetazone).

g Only available in combination with amoxicillin as co-amoxycrav (e.g. 500 mg amoxicillin/125 mg clavulanic acid fixed dose combination). It is given with each dose of carbapenem, either as 125 mg bd or 125 mg 3 times daily. h Tablets are expected to become available in the near future.

See the text of the [guidelines](#) for more details on the use of medicines.

For the BPaL regimen, the dosage of the drugs used in the NIX-TB trial was as follows: Bdq 400 mg QD for 2 weeks, followed by 200mg 3 times per week for a total of 26 weeks + Pa 200mg QD for 26 weeks + Lz 1200mg daily for 26 weeks

Annex 3. Dosing of medicines used in second-line MDR/RR-TB regimens by weight band in patients under 15 years

Source: 2019 WHO consolidated guidelines on drug-resistant tuberculosis treatment

Group	Medicine	Weightbased daily dose ^b	Formulation	Weight bands among patients not yet 15 years old ^a							Usual upper daily dose ^b	Comments
				5–6 kg	7–9 kg	10–15 kg	16–23 kg	24–30 kg	31–34 kg	>34 kg		
A	Levofloxacin	15–20 mg/kg	100 mg dt	1	1.5	2 or 3	3 or 4	(>14 y)	(>14 y)	(>14 y)	1.5 g	
			250 mg tab	0.5	0.5	1 or 1.5	1.5 or 2	2	3	(>14 y)	1.5 g	
	Moxifloxacin	10–15 mg/kg	100 mg dt	0.8	1.5	2	3	4	(>14 y)	(>14 y)	400 mg	
			400 mg tab ^c	2 ml ^c	3 ml ^c	5 ml ^c	0.5 or 0.75	1	(>14 y)	(>14 y)	400 mg	Use 10 mg/kg in <6 months
	Bedaquiline	-	100 mg tab	- ⁱ	- ⁱ	- ⁱ	2 tabs od for two weeks; then 1 tab od M/W/F for 22 weeks		4 tabs od for 2 weeks; then 2 tabs od M/W/F for 22 weeks		-	Only in patients >5 years old (lower dose from 15– 29 kg; higher dose from >29 kg)
	Linezolid	15 mg/kg od in <16 kg 10–12 mg/kg od in >15 kg	20 mg /ml susp	4 ml	6 ml	8 ml	11 ml	14 ml	15 ml	20 ml ^d		
			600 mg tab ^c	0.25	0.25	0.25	0.5	0.5	0.5	0.75 ^d	600 mg	
B	Clofazimine	2–5 mg/kg	50 mg cap or tab ^l	1 alt days	1 alt days	1 alt days	1	2	2	(>14 y)	100 mg	Give on alternate days if dose in mg/kg/day is too high
			100 mg cap or tab ^l	M/W/F	M/W/F	1 alt days	1 alt days	1	(>14 y)	(>14 y)	100 mg	

	Cycloserine or terizidone	15–20 mg/kg	125 mg mini capsule (cycloserine) ^c	1	1	2	3	4	(>14 y)	(>14 y)	1 g	
			250 mg cap ^c	4–5 ml ^c	5–6 ml ^c	7–10 ml ^c	2	2	2	(>14 y)	1 g	
C	Ethambutol	15–25 mg/kg	100 mg dt	1	2	3	4	-	-	(>14 y)	-	
			400 mg tab ^c	3 ml ^c	4 ml ^c	6 ml ^c	1	1 or 1.5	2	(>14 y)	-	
	Delamanid	-	50 mg tab	- ⁱ	-m, i	-em i	-m, i	1 bd ^h	1 bd ^h	2 bd ^m	200 mg	Only in patients >2 years old (25 mg bd in 3–5 years; 50 mg bd in 6–11 years; 100 mg bd in 12–17 years)
	Pyrazinamide	30–40 mg/kg	150 mg dt	1	2	3	4 or 5	-	-	(>14 y)	-	
			400 mg tab	0.5	0.75	1	1.5 or 2	2.5	3	(>14 y)	-	
			500 mg tab	0.5	0.5	0.75 or 1	1.5	2	2.5	(>14 y)	-	
	Imipenem-cilastatin	-	500 mg + 500 mg powder for injection, vial (10 ml)	-	-	-	-	-	-	-	-	Not used in patients <15 years (use meropenem)
	Meropenem	20–40 mg/kg iv every 8 hours	1 g powder for injection, vial (20 ml)	2 ml	4 ml	6 ml	8-9 ml	11 ml	(>14 y)	(>14 y)	-	To be used with clavulanic acid
	Amikacin	15–20 mg/kg	500 mg/2 ml solution for injection, ampoule ^f	0.4 ml	0.6 ml	0.8 - 1.0 ml	1.2 - 1.5 ml	2.0 ml	(>14 y)	(>14 y)	1 g	

	Streptomycin	20–40 mg/kg	1 g powder for injection, vial ^e	Calculate according to the dilution used						(>14 y)	(>14 y)	1 g	
	Ethionamide or prothionamide	15–20 mg/kg	125 mg dt (ethionamide)	1	1	2	3	4	4	(>14 y)		1 g	
			250 mg tab	0.5	0.5	1	2	2	2	(>14 y)		1 g	
	<i>p</i> -aminosalicylic acid	200–300 mg/kg in 2 divided doses	PAS acid (4 g) sachet	0.5–0.75 g bd	0.75–1 g bd	1–2 g bd	2–3 g bd	3–3.5 g bd	(>14 y)	(>14 y)	-		
			PAS sodium salt (equivalent to 4g PAS acid) sachet	0.5–0.75 g bd	0.75–1 g bd	1–2 g bd	2–3 g bd	3–3.5 g bd	(>14 y)	(>14 y)	-		
			PAS sodium salt 60% w/w (9.2 g; equivalent to 4g PAS acid)	1.5 g bd	2–3 g bd	3–4 g bd	4 or 6 g bd	6 or 8 g bd	8–12 g bd	8–12 g bd	-		
Other medicines ^f	Isoniazid	15–20 mg/kg (high dose)	50 mg/5 ml soln	8–10 ml	15 ml	20 ml	-	-	-	-			
			100 mg tab	1	1.5	2	3	4	4	(>14 y)	-		300 mg isoniazid tablet can be used in patients >20 kg Pyridoxine is always given with high-dose isoniazid in children (12.5 mg od in <5 y olds and 25 mg od in >4 y olds)

	Clavulanic acid ^g	-	62.5 mg clavulanic acid as amoxicillin/ clavulanate, 250 mg/62.5 mg, powder for oral solution, 5 ml	2 ml bd ^g	3 ml bd ^g	5 ml bd ^g	8 ml bd ^g	10 ml bd ^g	(>14 y)	(>14 y)	-	Only to be used with carbapenems
	Kanamycin	15–20 mg/kg	500 mg or 1 g powder for injection, vial (2 ml); ^e 1g/ 4 ml solution for	0.4 ml	0.6 ml	0.8–1.0 ml	1.2–1.5 ml	2.0 ml	(>14 y)	(>14 y)	1 g	1 g vials (3 ml) also available
			injection, ampoule ^e									
	Capreomycin	15–20 mg/kg	500 mg or 1 g powder for injection, vial (2 ml) ^e	0.4 ml	0.6 ml	0.8–1.0 ml	1.2–1.5 ml	2.0 ml	(>14 y)	(>14 y)	1 g	
	Gatifloxacin	-	400 mg tab	-	-	-	-	-	-	-	-	Currently there is no availability of gatifloxacin on the market. Gatifloxacin is not used in persons <18 years old.
	Thioacetazone	-	-	-	-	-	-	-	-	-	-	Currently there is no availability of thioacetazone on the market. Thioacetazone is not used in persons <18 years old.

(<15 y) = follow the separate dose schedule for patients younger than 15 years of age; bd = two times a day; cap = capsule; g = gram; im = intramuscular; iv = intravenous; kg = kilogram; ml = millilitre; mg = milligram; M/W/F = Monday, Wednesday, Friday; soln = solution; susp = suspension; tab = tablet

Footnotes

- a Dosages were established by the Guideline Development Group for the *WHO treatment guidelines for rifampicin- and multidrug-resistant tuberculosis, 2018 update* and the WHO Global task force on the pharmacokinetics and pharmacodynamics (PK/PD) of TB medicines and other experts. They are based on the most recent reviews and best practices in the treatment of MDR/RR-TB. For certain agents the dosages were informed by pharmacokinetic modelling results based on the principle of allometric scaling (Anderson BJ, Holford NH. Mechanism-based concepts of size and maturity in pharmacokinetics. *Annu Rev Pharmacol Toxicol* 2008;48:303–32). Due to the pharmacokinetic properties of certain medicines the doses proposed may exceed the mg/kg/day ranges shown here in order to achieve blood concentrations similar to target levels in an average adult patient. In patients <30kg kg follow the schedule for <15 year olds unless otherwise indicated. If multiple dose options are given for one weight band select the lower or higher option depending on whether the patient is at the lower or higher limit of the body weight range. Dosing more closely to the target mg/kg/day should be aimed for, and is more feasible with oral or parenteral fluids and when solid forms of different dosages are available. Fractioning of tablets into halves or less should be avoided, if possible. Therapeutic drug monitoring is advised when the dose is at the upper and lower ends of the range to minimize the adverse therapeutic consequences of over- and under-exposure, respectively (especially for injectable agents, linezolid and fluoroquinolones).
- b Clinicians may decide to exceed these values in particular cases to improve therapeutic effect.
- c No weight-based dosing is proposed.
- d Unless there is risk of toxicity, the high dose may be used if antimicrobial levels may be lowered because of pharmacokinetic interactions, malabsorption or other metabolic reasons or if the strain has low-level drug resistance.
- e Weight-based daily dose is for 6 or 7 days/week administration (M/W/F scheduling may permit higher dosing). Volumes shown may differ by preparation. Streptomycin may be diluted in three different ways. For iv use, the volume may be increased.
- f In the 2018 WHO treatment guidelines, these agents are either no longer recommended (kanamycin, capreomycin), only recommended as a companion agent (amoxicillin/clavulanic acid) or not included because of lack of data from the latest analysis on longer MDR/RR-TB regimens in adults (gatifloxacin, isoniazid and thioacetazone).
- g Only available in combination with amoxicillin as co-amoxycrav (e.g. 500 mg amoxicillin/125 mg clavulanic acid fixed dose combination). It is given with each dose of carbapenem, either as 125 mg bd or 125 mg 3 times daily.
- h *If using 25mg distab, dosing should be two tabs twice daily. Note that delamanid is not commissioned for those <3 years in the UK.*
- i *See Annex 4 of the guidelines*
- j *Tablets are expected to become available in the near future*
- k *May be used in children 3–5 years of age. Giving half a 50 mg adult tablet in these children does not result in the same blood levels observed in trials using the special 25 mg paediatric tablet. Bioavailability may further be altered when the 50 mg tablet is split, crushed or dissolved’*
See the text of the [guidelines](#) for more details on the use of medicines
- m *This would be 1 tablet BD according to UK commissioning groups. Delamanid is not commissioned for those <3 years or <30kg in the UK.*

Annex 4. Drugs and dosing in paediatric patients

Provisional, suggested doses of delamanid and bedaquiline for operational research for paediatric patients:

1) Bedaquiline

Weight range (kg)	Loading dose /day (mg) (first two weeks)	Maintenance dose mg/week (M/W/F)
3-7	100	150 (50 - 50 – 50)
>7 – 15	200	300 (100 –100 – 100)

kg = kilogram; mg = milligram; M/W/F = Monday, Wednesday, Friday

The bedaquiline “crush study” in adult volunteers* showed that bioavailability of bedaquiline 100 mg tabs suspended in water is equivalent to that for tablets swallowed whole. The suspension was well tolerated. These findings suggest that the currently available bedaquiline formulation (suspended in water) could be used to treat DR-TB in children, to bridge the gap until paediatric dispersible formulation becomes routinely available.

* Svensson et al; Br J Clin Pharmacol (2018) 84 2384–2392

2) Delamanid

Please note that the following dosing applies to the use of the 25mg dispersible tablet.

Please note that Delamanid is **not** currently commissioned for children <3 years or <30kg in the UK.¹

<12 kg		>12-23 kg+
Dissolve 1 tab of 25mg in 10 mL of water*, then give:		
5.5-8 kg	2 mL once daily	
>8-10 kg	2 mL twice daily	
>10-12 kg	4 mL twice daily	25 mg twice daily (using 25mg distab: 1 tab twice daily)

* Tablet needs to be fully dispersed before giving the aliquot, unused solution should not be conserved.

Child-friendly formulations available through the Global Drug Facility (GDF):

Pyrazinamide 150 mg *

Ethionamide 125 mg * Levofloxacin 100 mg *

Moxifloxacin 100 mg *

Cycloserine 125mg minicapsule Ethambutol 100 mg *

Isoniazid 100 mg *

* Dispersible tablet: to be dissolved in water and administered to the child as a solution

¹ NHS England. (17 July 2019). Clinical commissioning policy statement: treatment for defined patients with MDR-TB and XDR-TB including bedaquiline and delamanid.

<https://www.england.nhs.uk/commissioning/wp-content/uploads/sites/12/2019/07/clinical-commissioning-policy-statement-treatment-for-defined-patients-with-mdr-tb-and-xdr-updated-dec-20.pdf>

Annex 7. Inpatient and ambulatory treatment

Inpatient treatment is not mandatory, but patients may require hospitalisation at the initiation of MDR/RR-TB treatment for clinical reasons, or to ensure that patients can tolerate the regimen. It may also be advisable for specific groups and for very ill people, for instance during the initiation of treatment or when adverse events occur during treatment.

Directly Observed Therapy (DOT) (or VOT, video-observed therapy) should be administered throughout the whole treatment course and the shorter regimen should be administered seven days per week. Ambulatory DOT services could be either "facility-based" in which patients visit a health care facility daily for treatment, or "community based" in which a trained treatment supporter visits the patients daily for drug administration (or vice versa) and accompanies the patient to follow-up visits and liaises with the clinical staff. Enablers and incentives (such as travel expenditure or food) during the whole treatment course are helpful and should be consistently provided whenever possible and relevant to the local context.

In the case of community-based DOT, a trained independent treatment supporter who is not directly related to the patient must be identified. The treatment supporter has the following responsibilities:

- Administer DOT on a daily basis.
- Supports the patient to attend scheduled follow-up visits and examinations.
- Monitor adverse events closely and address adverse events in a timely manner by informing clinical staff.
- Update the patient treatment card on a daily basis.
- Initiate patient tracing if the patient fails to return for treatment as per schedule.
- Ensure that there is a sufficient buffer stock of drugs for patients who are currently on treatment.

Annex 8. Adverse events of special interest

All adverse events of special interest:

- Peripheral neuropathy (paraesthesia)
- Psychiatric disorders and central nervous system toxicity (e.g. depression, psychosis, suicidal intention, seizures)
- Optic nerve disorder (optic neuritis) or retinopathy
- Ototoxicity (hearing impairment, hearing loss)
- Myelosuppression (manifested as anaemia, thrombocytopenia, neutropenia or leukopenia)
- Prolonged QT interval (Fridericia correction)
- Lactic acidosis
- Hepatitis (defined as increases in alanine aminotransferase (ALT) or aspartate aminotransferase (AST) ≥ 5 x the upper limit of normal (ULN), or increases in ALT or AST ≥ 3 x ULN with clinical manifestations, or increases in ALT or AST ≥ 3 x ULN with concomitant increase in bilirubin ≥ 1.5 x ULN)
- Hypothyroidism
- Hypokalaemia
- Pancreatitis
- Phospholipidosis
- Acute kidney injury (acute renal failure)

Source: World Health Organization, Active tuberculosis drug-safety monitoring and management (aDSM), 2015

Annex 9. Adverse Events: grading, attribution, definitions

Table 1: Grading of Adverse Events

Source: DAIDS table for grading severity of adult and paediatric adverse events – corrected version 2.1 July 2017

Grade 1	Mild	Small or transient inconvenience that does not limit normal daily activity. No need for medical intervention or corrective treatment.
Grade 2	Moderate	Partial limitation of normal daily activity. In some, but not all cases, medical intervention or corrective treatment is necessary. No need to discontinue the treatment.
Grade 3	Severe	Limitation of normal daily activity. Medical intervention and corrective treatment, often requiring hospitalization, are necessary. The responsible drug may have to be stopped temporarily until the symptoms have resolved.
Grade 4	Life Threatening	Severe limitation of normal daily activity. Medical intervention and corrective treatment, requiring hospitalization, are necessary. The responsible drug may need to be stopped permanently.
Grade 5	Death	

*The full DAIDS table can be found at this web link:

<https://rsc.niaid.nih.gov/sites/default/files/daidsgradingcorrectedv21.pdf>

Table 2: Scale for attribution of assessing drug relationship with AE occurrence

Source: WHO TDR Workbook for investigators, 2012

<https://www.who.int/tdr/publications/documents/investigator.pdf>

Category	Definition
Definite	Events occurring within a timely manner after administration of the drug(s); that are known sequela to the administration of the drug(s) and follow a previously documented pattern of reaction, but for which no other explanation is known. This category applies to ADRs that the PI believes are incontrovertibly related to the treatment.
Probable	Any event occurring in a timely manner after administration of the drug(s); that follows a known pattern of reaction to the drug(s); and for which no other explanation is known. This category applies to ADRs that, after careful medical consideration at the time they are evaluated, are believed with a high degree of certainty to be related to the drug(s).
Possible	Any event occurring in a timely manner after administration of the drug(s) that does not follow a known pattern of reaction and for which no other explanation is known. This category applies to ADRs that, after careful medical consideration at the time they are evaluated, are considered unlikely to be related but that cannot be ruled out with certainty.
Unlikely	In general, this category can be considered applicable to those ADRs that, after careful medical consideration at the time they are evaluated, are considered to be unrelated to administration of the drug(s).

Not related	Any ADRs for which there is evidence that an alternative aetiology exists or for which no timely relationship exists to the administration of the drug(s) and the ADRs does not follow any previously documented pattern. This category applies to those ADRs that, after careful medical consideration, are clearly and incontrovertibly due to causes other than the drug(s).
Insufficient data to assess	There is insufficient information about the ADRs to allow for an assessment of causality.

Table 3: Definitions of Anticipated and Unanticipated Adverse Events

Source: ICH E2

Category	Definition
Anticipated	AE which meets any one (or both) of the following: 1. Consistent with listing in the current study protocol narrative and/or consent form and/or pharmaceutical package insert for the study treatment(s), in terms of the nature, intensity, and frequency. 2. Based on clinical judgment by the PI/primary treating physician, the AE was expected to occur based on the study participant's underlying medical condition and/or concomitant disorders, disease process, or non-study treatment.
Unanticipated	AE which meets BOTH of the following: 1. NOT consistent with listing in the current study protocol narrative and/or consent form and/or pharmaceutical package insert for the study treatment(s) in terms of the nature, intensity, and frequency. AND 2. Based on clinical judgment by the PI/primary treating physician, the AE was NOT expected to occur based on the study participant's underlying medical condition and/or concomitant disorder, disease process, or non-study treatment.

Annex 10. Modified Medical Research Council Dyspnea Scale

Source: Doherty DE et al. COPD: Consensus Recommendations for early diagnosis and treatment. *Journal of Family Practice*, Nov 2006

0	"I only get breathless with a strenuous exercise"
1	"I get short of breath when hurrying on the level or walking up a slight hill"
2	"I walk slower than people of the same age on the level because of breathlessness or have to stop for breath when walking at my own pace on the level"
3	"I stop for breath after walking about 100 yards or after a few minutes on the level"
4	"I am too breathless to leave the house" or "I am breathless when dressing"

Annex 11. Participant Information Sheet (PIS) and Informed Consent Form (ICF) – Adult participant

Part I: Information sheet

Title of study: TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom

Principal Investigator (PI):

Professor David Moore, London School of Hygiene and Tropical Medicine

Site Lead and contact details:

.....
.....
.....

Dear Sir/Madam,

You have been invited to take part in a research study because you have been diagnosed with tuberculosis and require a special course of treatment. This is either because your tuberculosis is resistant to the usual treatments or because you have not been able to tolerate the usual medications due to side effects.

We are conducting a study to look at the effect of different treatment courses for TB in the U.K. Please take some time to read the information presented here, which will explain the details of this study. Please ask us any questions about any part of the study that you do not fully understand. It is very important that you are fully satisfied that you clearly understand what this research entails and how you could be involved. Your participation is **entirely voluntary**, and **you are free to decline to participate**. If you say no, this will not negatively affect you or the quality of care provided to you in any way, including health care now and in the future. If you say yes, you will also be **free to withdraw from the study at any point** without having to give reasons for your withdrawal.

Ethics approval for this study was obtained from *[to be confirmed]*, and from *[to be confirmed]*. This study will be conducted according to ethical guidelines and principles of the International Declaration of Helsinki, as well as local ethical guidelines.

This study is being funded by *[to be confirmed pending application outcome]*.

What is this study all about?

Tuberculosis can be a difficult disease to treat. This study aims to understand better the different treatment options in the UK, specifically in cases where patients need special courses of treatment.

There are several reasons why people may need special courses of treatment. Your medical team will explain to you which of the following categories you fit into.

Multi-drug resistant TB:

Multi-drug resistant TB (MDR-TB) is TB that is resistant to at least two of the drugs used in the usual treatment course (rifampicin and isoniazid). For patients with MDR-TB, the standard treatment in the UK used to last 20-24 months and included drugs given by injection. This treatment course has high rates of side effects and may be unnecessarily long. More recently, shorter treatment courses (6-11 months) without injections have been recommended and are becoming more widely used internationally. There is good evidence from other countries that these treatment courses are effective and tolerated better but this has yet to be demonstrated in the UK. Doctors in the UK are starting to use shorter treatments in some cases, and we wish to study carefully the different range of treatment courses being used to look at their success rates, the rates of side effects and the impact on people's quality of life.

Isoniazid-resistant TB:

Some people have TB which is resistant to only one drug (isoniazid). Treatments for this condition can vary between centres in the UK and we would like to understand better what different centres are doing and how effective their treatments are.

Drug-intolerance:

Some people have TB which is **not** resistant but they are not able to tolerate the usual TB drugs due to side effects. These people also require treatment with different medications and, again, we would like to learn more about how this affects them and their care.

Why have I been invited to participate?

We are inviting all adults receiving special TB treatment courses in the UK to participate in this study.

What will happen to me if I take part?

If you accept to be part of the study, you will sign a consent form and an identification number (code) will be assigned to you. This will be used to anonymise the information that we collect about you.

Information that is routinely collected about you by the clinical team will then be recorded on a central database using your identification code. This database will be analysed throughout and at the end of the study period.

This information collected would include:

- Demographic data such as your age and gender
- Any comorbidities (other medical conditions that you might have)
- Clinical information about your TB diagnosis
- Clinical examination findings
- Your test results
- Your treatments and any side effects

To preserve confidentiality, none of the information in the database will be identified with your name, but only with the code assigned to you if you agree to participate. If you feel uncomfortable providing some sensitive information about you, please feel free to discuss it with the medical staff.

Taking part in the study will **not** affect the treatments given to you. Your clinical team will continue to treat you as per their usual practice, which is based on national guidelines, their clinical expertise and in some cases after discussions with an expert national panel. The purpose of the study is to collect information about you and your treatments so that we can understand better how these are working in the UK. If you choose to take part in the study, or if you choose **not** to take part in the study, the treatments given to you will be the same.

Since there is a possibility for your disease to reappear after treatment, you will also be asked to attend clinic appointments for up to one year after completing treatment. This will allow checking for any recurrence of the disease and appropriate clinical measures will be taken by the clinical team based on the results. This is standard practice for patients who have been treated for TB and attending these appointments is an important part of the care provided to you, regardless of whether you wish to take part in the study.

What do I have to do?

There is no difference for you regarding the quality of clinical care you will receive. You will come to the clinic to have medical consultations and to receive the medicines according to your clinical team's usual practice. Your clinic visits and tests will only consist of those that you would do anyway as part of the regular follow-up for patients on special treatment courses for tuberculosis.

You may be asked to fill out a questionnaire at some appointments to describe your experience of being treated for TB, in particular about your quality of life and any stigma (negative comments) that you may receive. The questionnaires will be given to you in clinic and can be filled out whilst at the clinic. You should ask your nurse or doctor about any concerns you have from these questionnaires. You do **not** have to fill out questionnaires and you can opt out of this part of the study on the consent form.

How will the data that we collect be treated?

Your name will **not** be written on the study collection forms, nor will it be included with the information entered into the electronic database. A number (code) will be used to identify your data instead. The electronic database will be hosted on servers of the World Health Organisation (WHO), who created the database to support individual countries in carrying out this research. Your data, in its anonymised format, will be visible to your own TB team in your hospital and to the central study team. Any information collected from you in hard-copy, for example your consent form and questionnaire responses, will be kept separate and will be locked away safely.

Your name will **not** be included in any of the reports or communication related to the study (including the publication of the findings).

Your routine test results (blood and sputum) will be recorded in the database; however, there will be no additional tests required and no change to the routine processing of these specimens done by your clinical team.

How will I benefit from taking part in this research?

This research will help us in preparing better treatment courses and protocols for patients with drug-resistant tuberculosis in the future. It might help the wider community in the United Kingdom to have access to improved treatments, but this is not guaranteed. Entering your data systematically onto the electronic database may help your clinical team to keep an overview of your progress.

What monitoring tests will I need?

You will need the same monitoring that is needed by all patients taking special courses of treatment for tuberculosis; this includes a first visit and a visit in 2 weeks' time followed by monthly visits. Participating in the study will not require any additional tests or clinic visits. Some tests may be requested from you such as blood tests, vision tests and tests to check your heart (electrocardiogram) in order to follow-up on possible side effects depending on what treatments your doctors think you need. You will be informed about the signs that should trigger you to see your doctor if you experience them.

Are there any risks involved in this research?

All the different procedures will be performed according to the framework of the recommended medical protocol, which has received approval from *[to be completed]*..... The master study protocol has also been endorsed by the World Health Organisation (WHO). Participating in this research will **not** affect the treatments given to you. Your care will continue to be provided as per the usual practice of your treating centre. The only risks associated with participation might relate to data security, but your research and clinical teams will make sure that your data is kept confidential with no identification with your name. All paper data related to the study will be kept locked and all electronic data will be protected by a password. In all cases, information related to the study will be only accessible to your usual TB team and the study team.

Do I have the right to refuse to participate or to withdraw from the study?

You do **not** have to agree to take part in this study if you do not wish to do so. Your participation is entirely voluntary, and you are free to decline to participate. If you say no, this will not affect you negatively in any way, including your treatment now or in the future. You are also free to withdraw from the study at any point, even if you do agree to take part now. If you decide to withdraw from the study, or if you are no longer able to continue consenting to your involvement, the data that was collected about you up to that point will still be used for analysis in an anonymised format and only for the purpose of this study; no new data will be collected from that point on.

Are there any costs involved and will I be paid to take part in this study?

There are no additional costs involved in taking part in this study. There is no payment for taking part.

If I have any other questions about the research in the future?

If you have any questions, you may contact the following person(s):

.....
.....

Part II: Informed Consent Form (ICF) for adult participant

IRAS ID: 317172

Centre Number:

Study Number:

Participant Identification Number for this trial:

CONSENT FORM

Title of Project: TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom

Name of Researcher:

Please initial box

1. I confirm that I have read the information sheet dated November 2023 (version 1.1) for the above study. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily. ☐
2. I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason, without my medical care or legal rights being affected. ☐
3. I understand that relevant sections of my medical notes and data collected during the study, may be looked at by individuals from the NHS Trust and study team, where it is relevant to my taking part in this research. I give permission for these individuals to have access to my records. ☐
4. I understand that the information collected about me may be used to support other research in the future, and may be shared anonymously with other researchers. ☐
5. I agree to my General Practitioner being informed of my participation in the study. ☐
6. I agree to take part in the above study. ☐

Optional extras (not required for participation in the main study):

7. I am happy to be given questionnaires to fill out at some appointments regarding my quality of life and any stigma (negative comments) I may receive. ☐
8. I am happy to be contacted in future about similar or related research projects ☐
9. I would like to be informed of the results of this study when complete (please provide your email address here if so Email address: _____) ☐

_____	_____	_____
Name of Participant	Date	Signature

_____	_____	_____
Name of person taking consent	Date	Signature

Annex 12. Participant Information Sheet (PIS) and Certificate of Assent – Child participant

This assent form is for children between the age of 6 and 18 and who have been selected to take part in this study and are able to read.

Part I: Information sheet

Title of study: TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom

Principal Investigator (PI):

Professor David Moore, London School of Hygiene and Tropical Medicine

Site Lead and contact details:

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.....

You have been invited to take part in a research study because you have been diagnosed with tuberculosis and require a special course of treatment. This is either because your tuberculosis is resistant to the usual treatments or because you have not been able to tolerate the usual medications due to side effects.

We are conducting a study to look at the effect of different treatment courses for TB in the U.K. Please take some time to read the information presented here, which will explain the details of this study. Please ask us any questions about any part of the study that you do not fully understand. It is very important that you are fully satisfied that you clearly understand what this research entails and how you could be involved. Your participation is **entirely voluntary**, and **you are free to decline to participate**. If you say no, this will not negatively affect you or the quality of care provided to you in any way, including health care now and in the future. If you say yes, you will also be **free to withdraw from the study at any point** without having to give reasons for your withdrawal.

Ethics approval for this study was obtained from *[to be confirmed]*, and from *[to be confirmed]*. This study will be conducted according to ethical guidelines and principles of the International Declaration of Helsinki, as well as local ethical guidelines.

This study is being funded by *[to be confirmed pending application outcome]*.

What is this study all about?

Tuberculosis can be a difficult disease to treat. This study aims to understand better the different treatment options in the UK, specifically in cases where patients need special courses of treatment.

There are several reasons why people may need special courses of treatment. Your medical team will explain to you which of the following categories you fit into.

Multi-drug resistant TB:

Multi-drug resistant TB (MDR-TB) is TB that is resistant to at least two of the drugs used in the usual treatment course (rifampicin and isoniazid). For patients with MDR-TB, the standard treatment in the UK used to last 20-24 months and included drugs given by injection. This treatment course has high rates of side effects and may be unnecessarily long. More recently, shorter treatment courses (6-11 months) without injections have been recommended and are becoming more widely used internationally. There is good evidence from other countries that these treatment courses are effective and tolerated better but this has yet to be demonstrated in the UK. Doctors in the UK are starting to use shorter treatments in some cases, and we wish to study carefully the different range of treatment courses being used to look at their success rates, the rates of side effects and the impact on people's quality of life.

Isoniazid-resistant TB:

Some people have TB which is resistant to only one drug (isoniazid). Treatments for this condition can vary between centres in the UK and we would like to understand better what different centres are doing and how effective their treatments are.

Drug-intolerance:

Some people have TB which is **not** resistant but they are not able to tolerate the usual TB drugs due to side effects. These people also require treatment with different medications and, again, we would like to learn more about how this affects them and their care.

Why have I been invited to participate?

We are inviting all adults receiving special TB treatment courses in the UK to participate in this study.

What will happen to me if I take part?

If you accept to be part of the study, you will sign a consent form and an identification number (code) will be assigned to you. This will be used to anonymise the information that we collect about you.

Information that is routinely collected about you by the clinical team will then be recorded on a central database using your identification code. This database will be analysed throughout and at the end of the study period.

This information collected would include:

- Demographic data such as your age and gender
- Any comorbidities (other medical conditions that you might have)
- Clinical information about your TB diagnosis
- Clinical examination findings
- Your test results
- Your treatments and any side effects

To preserve confidentiality, none of the information in the database will be identified with your name, but only with the code assigned to you if you agree to participate. If you feel uncomfortable providing some sensitive information about you, please feel free to discuss it with the medical staff.

Taking part in the study will **not** affect the treatments given to you. Your clinical team will continue to treat you as per their usual practice, which is based on national guidelines, their clinical expertise and in some cases after discussions with an expert national panel. The purpose of the study is to collect information about you and your treatments so that we can understand better how these are working in the UK. If you choose to take part in the study, or if you choose **not** to take part in the study, the treatments given to you will be the same.

Since there is a possibility for your disease to reappear after treatment, you will also be asked to attend clinic appointments for up to one year after completing treatment. This will allow checking for any recurrence of the disease and appropriate clinical measures will be taken by the clinical team based on the results. This is standard practice for patients who have been treated for TB and attending these appointments is an important part of the care provided to you, regardless of whether you wish to take part in the study.

What do I have to do?

There is no difference for you regarding the quality of clinical care you will receive. You will come to the clinic to have medical consultations and to receive the medicines according to your clinical team's usual practice. Your clinic visits and tests will only consist of those that you would do anyway as part of the regular follow-up for patients on special treatment courses for tuberculosis.

You may be asked to fill out a questionnaire at some appointments to describe your experience of being treated for TB, in particular about your quality of life and any stigma (negative comments) that you may receive. The questionnaires will be given to you in clinic and can be filled out whilst at the clinic. You should ask your nurse or doctor about any concerns you have from these questionnaires. You do **not** have to fill out questionnaires and you can opt out of this part of the study on the consent form.

How will the data that we collect be treated?

Your name will **not** be written on the study collection forms, nor will it be included with the information entered into the electronic database. A number (code) will be used to identify your data instead. The electronic database will be hosted on servers of the World Health Organisation (WHO), who created the database to support individual countries in carrying out this research. Your data, in its anonymised format, will be visible to your own TB team in your hospital and to the central study team. Any information collected from you in hard-copy, for example your consent form and questionnaire responses, will be kept separate and will be locked away safely.

Your name will **not** be included in any of the reports or communication related to the study (including the publication of the findings).

Your routine test results (blood and sputum) will be recorded in the database; however, there will be no additional tests required and no change to the routine processing of these specimens done by your clinical team.

How will I benefit from taking part in this research?

This research will help us in preparing better treatment courses and protocols for patients with drug-resistant tuberculosis in the future. It might help the wider community in the United Kingdom to have access to improved treatments, but this is not guaranteed. Entering your data systematically onto the electronic database may help your clinical team to keep an overview of your progress.

What monitoring tests will I need?

You will need the same monitoring that is needed by all patients taking special courses of treatment for tuberculosis; this includes a first visit and a visit in 2 weeks' time followed by monthly visits. Participating in the study will not require any additional tests or clinic visits. Some tests may be requested from you such as blood tests, vision tests and tests to check your heart (electrocardiogram) in order to follow-up on possible side effects depending on what treatments your doctors think you need. You will be informed about the signs that should trigger you to see your doctor if you experience them.

Are there any risks involved in this research?

All the different procedures will be performed according to the framework of the recommended medical protocol, which has received approval from *[to be completed]*..... The master study protocol has also been endorsed by the World Health Organisation (WHO). Participating in this research will **not** affect the treatments given to you. Your care will continue to be provided as per the usual practice of your treating centre. The only risks associated with participation might relate to data security, but your research and clinical teams will make sure that your data is kept confidential with no identification with your name. All paper data related to the study will be kept locked and all electronic data will be protected by a password. In all cases, information related to the study will be only accessible to your usual TB team and the study team.

Do I have the right to refuse to participate or to withdraw from the study?

You do **not** have to agree to take part in this study if you do not wish to do so. Your participation is entirely voluntary, and you are free to decline to participate. If you say no, this will not affect you negatively in any way, including your treatment now or in the future. You are also free to withdraw from the study at any point, even if you do agree to take part now. If you decide to withdraw from the study, or if you are no longer able to continue consenting to your involvement, the data that was collected about you up to that point will still be used for analysis in anonymised format and only for the purpose of this study; no new data will be collected from that point on.

Are there any costs involved and will I be paid to take part in this study?

There are no additional costs involved in taking part in this study. There is no payment for taking part.

If I have any other questions about the research in the future?

If you have any questions, you may contact the following person(s):

Part II: Certificate of Assent for Child

IRAS ID: 317172

Centre Number:

Study Number:

Participant Identification Number for this trial:

ASSENT FORM

Title of Project: TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom

Name of Researcher:

Please initial box

- | | |
|---|--------------------------|
| 1. I confirm that I have read the information sheet dated November 2023 (version 1.1) for the above study. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily. | ☐ |
| 2. I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason, without my medical care or legal rights being affected. | ☐ |
| 3. I understand that relevant sections of my medical notes and data collected during the study, may be looked at by individuals from the NHS Trust and study team, where it is relevant to my taking part in this research. I give permission for these individuals to have access to my records. | ☐ |
| 4. I understand that the information collected about me may be used to support other research in the future, and may be shared anonymously with other researchers. | ☐ |
| 5. I agree to my General Practitioner being informed of my participation in the study. | ☐ |
| 6. I agree to take part in the above study. | ☐ |

Optional extras (not required for participation in the main study):

- | | |
|--|--------------------------|
| 7. I am happy to be given questionnaires to fill out at some appointments regarding my quality of life and any stigma (negative comments) I may receive. | ☐ |
| 8. I am happy to be contacted in future about similar or related research projects | ☐ |
| 9. I would like to be informed of the results of this study when complete (please provide your email address here if so) Email address: _____ | ☐ |

_____	_____	_____
Name of Participant	Date	Signature

_____	_____	_____
Name of person taking consent	Date	Signature

Parent/Guardian has signed an informed consent: ☐ Yes ☐ No (initialled by researcher)

Annex 13. Participant Information Sheet (PIS) and Informed Consent Form – Parent / Guardian

Part I: Information sheet

Title of study: TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom

Principal Investigator (PI):

Professor David Moore, London School of Hygiene and Tropical Medicine

Site Lead and contact details:

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.....
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Dear Sir/Madam,

Your child has been invited to take part in a research study because they have been diagnosed with tuberculosis and require a special course of treatment. This is either because their tuberculosis is resistant to the usual treatments or because they have not been able to tolerate the usual medications due to side effects.

We are conducting a study to look at the effect of different treatment courses for TB in the U.K. Please take some time to read the information presented here, which will explain the details of this study. Please ask us any questions about any part of the study that you do not fully understand. It is very important that you are fully satisfied that you clearly understand what this research entails and how you and your child could be involved. Your participation is **entirely voluntary**, and **you are free to decline to participate**. If you say no, this will not negatively affect your child or the quality of care provided to them in any way, including health care now and in the future. If you say yes, you will also be **free to withdraw your child from the study at any point** without having to give reasons for this.

Ethics approval for this study was obtained from *[to be confirmed]*, and from *[to be confirmed]*. This study will be conducted according to ethical guidelines and principles of the International Declaration of Helsinki, as well as local ethical guidelines.

This study is being funded by *[to be confirmed pending application outcome]*.

What is this study all about?

Tuberculosis can be a difficult disease to treat. This study aims to understand better the different treatment options in the UK, specifically in cases where patients need special courses of treatment.

There are several reasons why people may need special courses of treatment. Your medical team will explain to you which of the following categories your child fits into.

Multi-drug resistant TB:

Multi-drug resistant TB (MDR-TB) is TB that is resistant to at least two of the drugs used in the usual treatment course (rifampicin and isoniazid). For patients with MDR-TB, the standard treatment in the UK used to last 20-24 months and included drugs given by injection. This treatment course has high rates of side effects and may be unnecessarily long. More recently, shorter treatment courses (6-11 months) without injections have been recommended and are becoming more widely used internationally. There is good evidence from other countries that these treatment courses are effective and tolerated better but this has yet to be demonstrated in the UK. Doctors in the UK are starting to use shorter treatments in some cases, and we wish to study carefully the different range of treatment courses being used to look at their success rates, the rates of side effects and the impact on people's quality of life.

Isoniazid-resistant TB:

Some people have TB which is resistant to only one drug (isoniazid). Treatments for this condition can vary between centres in the UK and we would like to understand better what different centres are doing and how effective their treatments are.

Drug-intolerance:

Some people have TB which is **not** resistant but they are not able to tolerate the usual TB drugs due to side effects. These people also require treatment with different medications and, again, we would like to learn more about how this affects them and their care.

Why has my child been invited to participate?

We are inviting all patients (children and adults) receiving special TB treatment courses in the UK to participate in this study.

What will happen to my child if they take part?

If you accept for your child to be part of this study, you will sign a consent form and an identification number (code) will be assigned to your child. This will be used to anonymise the information that we collect about them.

Information that is routinely collected about your child by the clinical team will then be recorded on a central database using their identification code. This database will be analysed throughout and at the end of the study period.

This information collected would include:

- Demographic data such as their age and gender
- Any comorbidities (other medical conditions that they might have)
- Clinical information about their TB diagnosis
- Clinical examination findings
- Their test results
- Their treatments and any side effects

To preserve confidentiality, none of the information in the database will be identified with your child's name, but only with the code assigned to them if you agree to participate. If you feel

uncomfortable providing some sensitive information about your child, please feel free to discuss it with the medical staff.

Taking part in the study will **not** affect the treatments given to your child. Their clinical team will continue to treat them as per their usual practice, which is based on national guidelines, their clinical expertise and in some cases after discussions with an expert national panel. The purpose of the study is to collect information about your child and their treatments so that we can understand better how these are working in the UK. If you choose to take part in the study, or if you choose **not** to take part in the study, the treatments given to your child will be the same.

Since there is a possibility for your child's disease to reappear after treatment, they will also be asked to attend clinic appointments for up to one year after completing treatment. This will allow checking for any recurrence of the disease and appropriate clinical measures will be taken by the clinical team based on the results. This is standard practice for patients who have been treated for TB and attending these appointments is an important part of the care provided to your child, regardless of whether you wish to take part in the study.

What do my child and I have to do?

There is no difference for you regarding the quality of clinical care your child will receive. Your child will come to the clinic to have medical consultations and to receive the medicines according to their clinical team's usual practice. Their clinic visits and tests will only consist of those that they would do anyway as part of the regular follow-up for patients on special treatment courses for tuberculosis.

They may be asked to fill out a questionnaire at some appointments to describe their experience of being treated for TB, in particular about their quality of life and any stigma (negative comments) that they may receive. The questionnaires will be given to you in clinic and can be filled out whilst at the clinic. You and your child should ask your nurse or doctor about any concerns you have from these questionnaires. Your child does **not** have to fill out questionnaires and you can opt out of this part of the study on the consent form.

How will the data that we collect be treated?

Your child's name will **not** be written on the study collection forms, nor will it be included with the information entered into the electronic database. A number (code) will be used to identify your child's data instead. The electronic database will be hosted on servers of the World Health Organisation (WHO), who created the database to support individual countries in carrying out this research. Your child's data, in its anonymised format, will be visible to their own TB team in their hospital and to the central study team. Any information collected from your child in hard-copy, for example the consent form and questionnaire responses, will be kept separate and will be locked away safely.

Your child's name will **not** be included in any of the reports or communication related to the study (including the publication of the findings).

Your child's routine test results (blood and sputum) will be recorded in the database; however, there will be no additional tests required and no change to the routine processing of these specimens done by the clinical team.

How will my child and I benefit from taking part in this research?

This research will help us in preparing better treatment courses and protocols for patients with drug-resistant tuberculosis in the future. It might help the wider community in the United Kingdom to have access to improved treatments, but this is not guaranteed. Entering your child's data systematically onto the electronic database may help their clinical team to keep an overview of their progress.

What monitoring tests will my child need?

Your child will need the same monitoring that is needed by all patients taking special courses of treatment for tuberculosis; this includes a first visit and a visit in 2 weeks' time followed by monthly visits. Participating in the study will not require any additional tests or clinic visits. Some tests may be requested from your child such as blood tests, vision tests and tests to check their heart (electrocardiogram) in order to follow-up on possible side effects depending on what treatments your doctors think your child needs. You and your child will be informed about the signs that should trigger you to see the doctor if your child experiences them.

Are there any risks involved in this research?

All the different procedures will be performed according to the framework of the recommended medical protocol, which has received approval from *[to be completed]*..... The master study protocol has also been endorsed by the World Health Organisation (WHO). Participating in this research will **not** affect the treatments given to your child. Their care will continue to be provided as per the usual practice of their treating centre. The only risks associated with participation might relate to data security, but the research and clinical teams will make sure that your child's data is kept confidential with no identification with their name. All paper data related to the study will be kept locked and all electronic data will be protected by a password. In all cases, information related to the study will be only accessible to your child's usual TB team and the study team.

Do I have the right to refuse to participate or to withdraw from the study?

You do **not** have to agree to take part in this study if you do not wish to do so. Your participation is entirely voluntary, and you and your child are free to decline to participate. If you say no, this will not affect your child negatively in any way, including their treatment now or in the future. You are also free to withdraw your child from the study at any point, even if you do agree to take part now. If you decide to withdraw them from the study, or if you are no longer able to continue consenting to their involvement, the data that was collected about your child up to that point will still be used for analysis in anonymised format and only for the purpose of this study; no new data will be collected from that point on.

Are there any costs involved and will I be paid to take part in this study?

There are no additional costs involved in taking part in this study. There is no payment for taking part.

If I have any other questions about the research in the future?

If you have any questions, you may contact the following person(s):

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Informed Consent Form (ICF) for parent / guardian

IRAS ID: 317172

Centre Number:

Study Number:

Participant Identification Number for this trial:

CONSENT FORM (Parent / Guardian)

Title of Project: TB ShORRT UK: an implementation evaluation of shorter, all-oral regimens for drug-resistant tuberculosis in the United Kingdom

Name of Researcher:

Please initial box

1. I confirm that I have read the information sheet dated November 2023 (version 1.1) for the above study. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily. ☐
2. I understand that my child's participation is voluntary and that we are free to withdraw at any time without giving any reason, without my child's medical care or legal rights being affected. ☐
3. I understand that relevant sections of my child's medical notes and data collected during the study, may be looked at by individuals from the NHS Trust and study team, where it is relevant to my taking part in this research. I give permission for these individuals to have access to my records. ☐
4. I understand that the information collected about my child may be used to support other research in the future, and may be shared anonymously with other researchers. ☐
5. I agree to my child's General Practitioner being informed of my participation in the study. ☐
6. I agree for my child to take part in the above study. ☐

Optional extras (not required for participation in the main study):

7. I am happy to be given questionnaires to fill out at some appointments regarding my child's quality of life and any stigma (negative comments) they may receive. ☐
8. I am happy to be contacted in future about similar or related research projects ☐
9. I would like to be informed of the results of this study when complete (please provide your email ☐

address here if so: Email address: _____

_____	_____	_____
Name of Parent / Guardian	Date	Signature

_____	_____	_____
Name of person taking consent	Date	Signature

